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Best of both worlds: Stochastic & adversarial best-arm identification

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Abstract

We study bandit best-arm identification with arbitrary and potentially adversarial rewards. A simple random uniform learner obtains the optimal rate of error in the adversarial scenario. However, this type of strategy is suboptimal when the rewards are sampled stochastically. Therefore, we ask: *Can we design a learner that performs optimally in both the stochastic and adversarial problems while not being aware of the nature of the rewards?* First, we show that designing such a learner is impossible in general. In particular, to be robust to adversarial rewards, we can only guarantee optimal rates of error on a subset of the stochastic problems. We introduce a new complexity measure, H_{BOB}, and establish a lower bound on the stochastic error rates achievable by learners that must also remain robust to adversarial rewards. Finally, we design a simple parameter-free algorithm and show that its probability of error matches (up to log factors) the lower bound in stochastic problems, and it is also robust to adversarial ones.

Keywords: multi-armed bandits, best-arm identification, adversarial and stochastic rewards

1. Introduction

In best-arm identification (Maron and Moore, 1993; Bubeck et al., 2009), the *learner* tries to identify the *arm* (option, decision) with the highest (expected) average *reward* among K given arms. At each round t of the *game* (the interaction of the learner with its environment), each arm k is assigned a reward $\boldsymbol{g}_{k,t}$. On this very same round t, a learner chooses an arm I_t and *only* observes the reward of that arm, $\boldsymbol{g}_{I_t,t}$, while the rest of the vector $\boldsymbol{g}_t$ is hidden from the learner.

Typically, we assume stochastic rewards: Each arm k is associated with a distribution ν_k and, for all k and t, the $\boldsymbol{g}_{k,t}$ is sampled i.i.d. from ν_k. In this paper, we aim for a robust solution for this setting and we allow the possibility that the rewards are non-stochastic. The rewards could have been chosen even by an oblivious *adversary*: The rewards are fixed before the start of the game but they are not necessarily drawn i.i.d. from a distribution. We focus on *fixed-budget*, where a total number

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of arm pulls n is fixed and the learner wishes to identify, as accurately as possible, the arm that attains the highest cumulative reward. We also discuss possible extensions to the fixed-confidence case, *adaptive* adversaries, and other settings in Section 6.

Given $\boldsymbol{g}$ and round n, we define the cumulative gain of arm k as $G_k = \sum_{t=1}^{n} \boldsymbol{g}_{k,t}$. Same as for adversarial bandits for cumulative-regret (Auer et al., 2002), the best arm in hindsight is defined as $k^\star_{\boldsymbol{g}} = \operatorname{argmax}_{k\in[K]} G_k$.[1] In a similar way, we define the gaps in hindsight with respect to $\boldsymbol{g}$ between two arms k and j as $\Delta^{\boldsymbol{g}}_{k,j} \triangleq \frac{1}{n}\sum_{t=1}^{n}(\boldsymbol{g}_{k,t} - \boldsymbol{g}_{j,t})$, giving a good proxy for the difficulty of discriminating between these two arms even in an adversarial environment. We design a learner and show that the probability[2] of error at round n *given* $\boldsymbol{g}$ against any fixed adversarial reward design $\boldsymbol{g}$ is bounded by a measure of complexity depending on the gaps in hindsight with respect to $\boldsymbol{g}$ and n.

Next, we discuss the motivations for studying non-stochastic best-arm identification. First, learning in the presence of adversarial data implies robustness. In a real-world best-arm identification, such as clinical trials or online ad recommendation, the assumption of i.i.d. data may not be valid. For instance, successive rewards from the same arm could be correlated. It is also possible that an adversary is trying to obscure the correct results: For example, the adversary might use a botnet to make the learner sell more ads. As discussed by Bubeck and Cesa-Bianchi (2012, Section 3), a deterministic learner or a learner that eliminates arms with low observed cumulative reward in an early stage of the game could be easily fooled by an adversary feeding it uninformative rewards in each of its deterministic pulls or in early stages of the game. Therefore, an efficient learner needs to employ internal randomization and pull each arm with a positive probability $\boldsymbol{p}_{k,t}$, conditional on the information available immediately before the action at round t.

Best-arm identification with completely adversarial rewards is so difficult that a very conservative approach is already near-optimal. In Section 3 we show that by playing the arms *uniformly at random*, the learner obtains the optimal gap-dependent rates of error against worst-case adversarial sequences. In the stochastic case, however, picking arms uniformly at random is suboptimal. This reveals the **best of both worlds (BOB)** question: *Is there a learner that attains the optimal error rates in both the stochastic and adversarial settings without knowledge of the environment?*

To study the above question, we face a number of **challenges**: How can a learner maintain the randomization and exploration needed for adversarial robustness while sampling the most promising arms more frequently? Can the learner detect if the rewards are stochastic or adversarial? What is an appropriate estimate of G_k? In the stochastic case, after arm k has been sampled at least once, $\widehat{G}_k \triangleq \frac{n\sum_{t=1}^{n}\mathbf{1}\{I_t=k\}\boldsymbol{g}_{k,t}}{\sum_{t'=1}^{n}\mathbf{1}\{I_{t'}=k\}}$ is commonly used, but if not used carefully, it can be easily biased by an adversary. Before the first sample, an initialization convention is required. In the adversarial case, $\widetilde{G}_k \triangleq \sum_{t=1}^{n}\frac{\boldsymbol{g}_{k,t}}{\boldsymbol{p}_{k,t}}\mathbf{1}\{I_t = k\}$ is usual but it can have a high variance if $\boldsymbol{p}_{k,t}$ is small (for rewards in $[0,1]$, the sum of its one-step conditional variances is bounded by $\sum_{t=1}^{n}(1/\boldsymbol{p}_{k,t})$). Controlling this potentially large variance, especially when dealing with a stochastic problem, is one of the main challenges that we face. In particular, high variance can happen if a learner explores uniformly for too long.

Our contributions We consider a new formulation of the adversarial best-arm identification. We study whether a BOB result is possible for fixed-budget best-arm identification. We answer the question negatively and show that for a class of stochastic problems, no robust (to adversary) learner

1. An alternative definition of best-arm identification could be predicting $\operatorname{argmax}_{k\in[K]}\boldsymbol{g}_{k,n+1}$. However, this is impossible in the adversarial case where $\boldsymbol{g}_{k,n+1}$ can be chosen arbitrarily without any dependence on $\boldsymbol{g}_{k,[n]}$.
2. Note that in our setup, given $\boldsymbol{g}$, the randomness comes solely from the potential internal randomization of the learner.

can achieve the optimal stochastic error rates. To prove this result, we introduce a new measure of complexity of the stochastic part of the task. This measure of complexity gives the problem-dependent error rates that a robust learner can guarantee in any stochastic problem. We study this new complexity in different stochastic regimes and provide several positive examples where BOB is possible but also several negative ones. We notice that even in stochastic problems where our lower bound seems to indicate that BOB is achievable, the robust uniform learner is clearly suboptimal. Therefore there is a need for a new algorithm to bridge the gap. In Section 5, we design a simple parameter-free learner and show that its error rate matches, up to log factors, the lower bound in the stochastic case as well as those of the uniform strategy in the adversarial case.

Formal verification The principal theoretical results of this paper have been formalized in Lean 4. The machine-checked proofs and verification instructions will be available upon public release at `https://github.com/victorgabillon/best-of-both-worlds-bandits`.

Related work The stochastic best-arm identification was introduced in the fixed-budget setting by Bubeck et al. (2009) and Audibert et al. (2010). Refined upper and lower bounds can be found respectively in the works of Karnin et al. (2013) and Carpentier and Locatelli (2016). For cumulative regret, the BOB question was raised by Bubeck and Slivkins (2012). Seldin and Slivkins (2014) gave a practical algorithm for addressing the same problem (see also Seldin and Lugosi, 2017). A lower bound and a refined upper bound for the problem were given by Auer and Chiang (2016).

Best-arm identification has been studied in a different non-stochastic setting by Jamieson and Talwalkar (2016) and Li et al. (2016). At round t for its m-th pull of arm k, their learner observes $\boldsymbol{g}_{k,m}$, whereas our learner observes $\boldsymbol{g}_{k,t}$. Moreover, their work is specifically tailored for online hyperparameter optimization of learning methods, where the value of each hyperparameter is assumed to converge, at some unknown rate, to its true value as it is given more resources (e.g., data or time). Therefore, their objective is to identify the best arm once the convergence has happened while in our setting, we do not assume any convergence and are interested instead in comparing the cumulative rewards G_k of each arm. By assuming convergence of the arms and asymptotically having the learner observe all the rewards, they prove that a state-of-the-art deterministic learner from the vanilla stochastic best-arm identification also solves their setting.

Allesiardo and Féraud (2017) and Allesiardo et al. (2017) analyze a non-stationary stochastic best-arm problem in a fixed-confidence setting but under the assumption that the game can be split into independent sub-games where the identity of the best arm does not change. This assumption precludes many of the hard examples where the adversary tricks the learner in the early stages of the game. The learner can then simply use a randomized version of Hoeffding Races and safely stop pulling arms. Also, while our notion of gaps is defined for round n, they define gaps at any intermediate round of the game but then consider the minimum gaps over time for each arm. This leads to a larger notion of complexity and permits ignoring the variance of the estimates.

Two types of applications In the first type, the learner believes that future rewards will be similar to the ones already observed. Here, from the n observations, the learner could use the estimated best arm in the upcoming rounds. In an expert setting, we might test the quality of experts for a limited time before committing to one. We may also optimize the hyperparameters of algorithms. Contrary to Jamieson and Talwalkar (2016), our objective is to find the arm (hyperparameter) with the highest cumulative value over a test set, rather than the performance that is achieved by the hyperparameter after convergence. In the second type, there are no new upcoming rounds. The identity of the best

arm is used to take an action based on the collected data. As an example, consider a law-enforcing agency that collects information periodically from different targets, one target per week, and at the end of the year, it decides which target to investigate thoroughly over its activities in the past year. This problem can be seen as a game between the agency and the malignant targets. Therefore, it would be useful to have algorithms that are robust to worst-case scenarios but still take advantage of easy data in case malignant targets actually do not take precautions to avoid being caught. Finally, in our setting, we are not trying to be robust to corruption of the data, as we want to find the best arm whether it includes corruption or not, unlike Altschuler et al. (2018) who study corrupted bandits.

2. Problem formulation for adversarial and stochastic rewards

Let $[a:b] = \{a, a+1, \dots, b\}$ with $a, b \in \mathbb{N}$, $a \leq b$, and $[a] = [1:a]$. A vector indexed by both round t and a specific element index k is $\boldsymbol{w}_{k,t}$. We detail the general game protocol in Figure 1.

Input: the time horizon n and the number of arms K
For $t = 1, 2, \dots, n$,
- simultaneously, the learner chooses arm $I_t \in [K]$, and
- the adversary picks a gain vector $\boldsymbol{g}_t \in [0,1]^K$.
- Then, the learner observes $\boldsymbol{g}_{I_t,t}$.

After round n is over, the learner recommends the arm J_n with the objective that $J_n = k^\star_{\boldsymbol{g}}$.

Figure 1: General protocol of the adversarial setting. The adversary is oblivious.

Adversarial case The adversary is denoted as ADV. It is the process that generates $\boldsymbol{g}$. Let (m) denote the index of the m-th best arm in $[K]$ and $G_{(m)}$ its corresponding cumulative gain so that $G_{(1)} > G_{(2)} \geq \ldots \geq G_{(K)}$. Dually to $(\cdot)$, $\langle k \rangle$ denotes the rank of arm k after sorting according to $G_{(\cdot)}$ so that $\langle (k) \rangle = (\langle k \rangle) = k$, $\forall k \in [K]$. Without loss of generality, note that we assumed there exists a unique best arm $k^\star_{\boldsymbol{g}} = (1)$. For each arm $k \in [K]$, we define the gap $\Delta^{\boldsymbol{g}}_k$ as

$$n\Delta^{\boldsymbol{g}}_k \triangleq \begin{cases} G_{(1)} - G_k & \text{if } k \neq k^\star_{\boldsymbol{g}}, \\ G_{(1)} - G_{(2)} & \text{if } k = k^\star_{\boldsymbol{g}}. \end{cases}$$

The gap can also be written as $n\Delta^{\boldsymbol{g}}_k = \left| \max_{i \neq k} G_i - G_k \right|$. $J_n \in [K]$ is the arm returned by the learner at the end of the exploration phase. Given a budget n and a fixed adversarial set of rewards $\boldsymbol{g}$ designed by an adversary ADV, the performance of the learner is measured by the probability $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$ of not identifying the best arm, i.e., $e_{\mathrm{ADV}(\boldsymbol{g})}(n) \triangleq \mathbb{P}\left(J_n \neq k^\star_{\boldsymbol{g}}\right)$. The smaller $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$, the better the learner. The probability is taken over the randomness of the learner as $\boldsymbol{g}$ is fixed. An alternative definition of the best arm, i.e., with highest G_k in expectation over adversarially sampled $\boldsymbol{g}$, would lead to an impossible problem. Indeed, it could happen that the best arm in expectation is actually always clearly suboptimal on any realization $\boldsymbol{g}$. We define the random variables $\widetilde{\boldsymbol{g}}_{k,t}$ as

$$\widetilde{\boldsymbol{g}}_{k,t} \triangleq \frac{\boldsymbol{g}_{k,t} \mathbf{1}\{I_t = k\}}{\boldsymbol{p}_{k,t}},$$

for arm k at round t and for which $\mathbb{E}_{I_t \sim \boldsymbol{p}_t}\left[\widetilde{\boldsymbol{g}}_{k,t}\right] = \boldsymbol{g}_{k,t}$. We similarly define $\widetilde{G}_{k,t} \triangleq \sum_{t'=1}^{t} \widetilde{\boldsymbol{g}}_{k,t'}$.

Stochastic case In the stochastic setting (Audibert et al., 2010), denoted STO, the distribution ν_k of arm k is bounded in $[0,1]$ with mean μ_k. The ordering $(\cdot)$ is such that $\mu_{(1)} > \mu_{(2)} \geq \ldots \geq \mu_{(K)}$, as we assume the uniqueness of the best arm without loss of generality. The distributions $\{\nu_k\}$ are unknown to the learner. The best arm to be identified is $k^\star_{\text{STO}} = \operatorname{argmax}_{k\in[K]}\mu_k$. Similar to the adversarial case, the gaps are $\Delta_k \triangleq |\max_{i\neq k}\mu_i - \mu_k|$, ranked as $\Delta_{(1)} \triangleq \Delta_{(2)} \leq \ldots \leq \Delta_{(K)}$, and the error of the learner is $e_{\text{STO}}(n) \triangleq \mathbb{P}\left(J_n \neq k^\star_{\text{STO}}\right)$. However, unlike in the adversarial case, this definition of the error includes the randomness of rewards. For this comparison, if several arms tie for the largest cumulative gain, let $k^\star_{\boldsymbol{g}}$ be the one with the smallest index. By Hoeffding's inequality and a union bound, a table $\boldsymbol{g} \sim$ STO satisfies $k^\star_{\text{STO}} \neq k^\star_{\boldsymbol{g}}$ with probability at most $2K\exp(-n\Delta^2_{(1)}/2)$. The two error indicators can differ only on this event, so the absolute difference between their probabilities, both averaged over the stochastic rewards and learner randomness, is bounded by this disagreement probability.

Notions of complexity Given gaps Δ_k for $k \in [K]$, we define two notions of complexity of the identification task, $H_{\texttt{SR}}$ and H_{UNIF}, that were introduced by Audibert et al. (2010). In particular,

$$H_{\texttt{SR}} \triangleq \max_{k\in[K]} \frac{k}{\Delta^2_{(k)}} \qquad \text{and} \qquad H_{\text{UNIF}} \triangleq \frac{K}{\Delta^2_{(1)}}.$$

$H_{\texttt{SR}}$ relates to the complexity of the stochastic case. H_{UNIF} will be used both in the adversarial and stochastic cases. In the adversarial case, the complexity and the gaps are defined with respect to $\boldsymbol{g}$, but we also sometimes write the uniform complexity as $H_{\text{UNIF}}(\boldsymbol{g})$ for clarity.

Class of problems We define a set of classes to group problems with very similar gap structure and with complexities that are only a constant multiplicative factor apart. For any $0 < \Delta_1 = \Delta_2 \leq \Delta_3 \leq \ldots \leq \Delta_K \leq 1/8$, we define a *problem class* $\boldsymbol{\Delta}_c$ with $c \geq 1$. Given these gaps and c, in the adversarial case, we say that $\boldsymbol{g} \in \boldsymbol{\Delta}_c$ if for all $k \in [K]$, $\Delta_k/c \leq \Delta^{\boldsymbol{g}}_k \leq c\Delta_k$ except for only one arm $\overline{k}$ whose gap is related to the smallest gap as $\Delta_1/c \leq \Delta^{\boldsymbol{g}}_{\overline{k}} \leq c\Delta_1$. In the stochastic case, STO $\in \boldsymbol{\Delta}_c$ if the same inequalities, including the exception for arm $\overline{k}$, hold with the actual gaps $|\max_{i\neq k}\mu_i - \mu_k|$ in place of $\Delta^{\boldsymbol{g}}_k$, while keeping the reference gaps Δ_k fixed.

The exceptional arm allows our lower-bound constructions to change the identity of the best arm while keeping the gaps of the remaining arms comparable to the same reference gap profile.

3. General adversarial case: An optimal learner can play uniformly at random

In this section, we define a simple learner (`Rule`) and in Theorem 1, we provide an upper bound on its probability of error depending on the gap in hindsight. We also give a matching lower bound for the general adversarial best-arm identification (Theorem 2) proving that `Rule` obtains the optimal gap-dependent rates of error against worst-case adversaries.

At round t, the *random uniform learner* (`Rule`) selects an arm $I_t \in [K]$ uniformly at random, i.e., with probability $\boldsymbol{p}_{k,t} \triangleq \mathbb{P}(I_t = k) \triangleq 1/K$ for all $k \in [K]$. Each round uses a fresh uniform draw, independent of past actions and observations. At the end of the game, the recommended arm J_n is the one with highest estimated cumulative gain, $J_n \triangleq \operatorname{argmax}_{k\in[K]}\widetilde{G}_{k,n}$.

Theorem 1 (Upper bound for `Rule` in the adversarial case) *For any horizon n, given rewards $\boldsymbol{g}$ chosen by an oblivious adversary, with $\boldsymbol{g}_{k,t} \in [0,1]$ for all $t \in [n]$ and for all $k \in [K]$,* `Rule` *outputs*

an arm J_n with the guarantee that its probability of error $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$ verifies

$$e_{\mathrm{ADV}(\boldsymbol{g})}(n) \leq K \exp\left(-\frac{3n}{28 H_{\mathrm{UNIF}}(\boldsymbol{g})}\right).$$

Sketch of the proof *(full proof in Appendix B)*: The deviation of $\widetilde{G}_{k,t}$ from $G_{k,t}$ is bounded by a Bernstein bound: for each fixed arm k, given the fixed table $\boldsymbol{g}$, the $\widetilde{\boldsymbol{g}}_{k,t}$ are independent across rounds because sampling uses fresh uniform draws. They are scaled Bernoulli$(1/K)$ indicators with variance at most K. Bernstein uses this variance bound of order K, whereas Hoeffding would use the squared range of order K^2. A union bound combines one lower tail for the best arm with one upper tail for each competitor.

Theorem 2 (Lower bound for the adversarial problem) *Given any problem class $\boldsymbol{\Delta}_3$ with associated complexity H_{UNIF}, for any learner, for any horizon n such that $K \exp\left(-n\Delta_1^2/128\right) \leq 1/128$ and $K \geq 4096$, there exist $\boldsymbol{g}^1 \in \boldsymbol{\Delta}_3$ and $\boldsymbol{g}^2 \in \boldsymbol{\Delta}_3$ so that the probabilities of error suffered by the learner on $\boldsymbol{g}^1$ and $\boldsymbol{g}^2$, denoted $e_{\boldsymbol{g}^1}(n)$ and $e_{\boldsymbol{g}^2}(n)$ respectively, verify*

$$\max\left(e_{\boldsymbol{g}^1}(n), e_{\boldsymbol{g}^2}(n)\right) \geq \min\left(\frac{1}{128}\exp\left(-\frac{32n}{H_{\mathrm{UNIF}}}\right), \frac{1}{32}\right).$$

Sketch of the proof *(full proof given in Appendix D)*: The key idea is to identify an arm that the learner samples infrequently and use it to construct two environments that are difficult to distinguish from the learner's observations but have different best arms with high probability. Consider a fixed learner run with horizon n, and a base problem BASE with Bernoulli means $\mu_1^{\mathrm{BASE}} = 1/2$ and $\mu_j^{\mathrm{BASE}} = 1/2 - \Delta_j$ for $j \in [2:K]$. During the first $m = \lfloor n/2 \rfloor$ rounds, Dirichlet's box principle gives a fixed suboptimal arm $\overline{k}$ whose expected number of pulls is at most $m/(K-1) \leq n/K$. We select this arm using expectations under BASE, before drawing the reward tables, and use it to construct two similar problems.

In the first phase of m rounds, ADV_1 follows BASE on every arm. The problem ADV_2 differs only in arm $\overline{k}$, whose Bernoulli mean is $1/2 - \Delta_{\overline{k}} + 2\Delta_1$ rather than $1/2 - \Delta_{\overline{k}}$. In both problems, the second phase also has m rounds, from $m+1$ through $2m$: arm $\overline{k}$ receives the deterministic reward $\Delta_{\overline{k}} - \Delta_1$ in each round, and all other arms receive zero. If the horizon is odd, the remaining final round gives zero reward to every arm. The complete table is drawn before interaction, and the learner still acts for all n rounds and recommends an arm after the last round.

The common deterministic second phase makes arm $\overline{k}$ a close competitor to arm 1, irrespective of its original gap. In the two problems, the expected cumulative gains satisfy

$$\begin{aligned}\mathbb{E}_1[G_1] = \mathbb{E}_2[G_1] &= m/2,\\ \mathbb{E}_1[G_{\overline{k}}] &= m(1/2 - \Delta_1),\\ \mathbb{E}_2[G_{\overline{k}}] &= m(1/2 + \Delta_1).\end{aligned}$$

Thus the selected arm lies just below arm 1 in expected total reward in the first problem and just above it in the second.

Only first-phase pulls of arm $\overline{k}$ distinguish the problems. Although its reward laws differ during m rounds, the learner obtains at most n/K such observations in expectation under ADV_1, whose prefix agrees with BASE. Since the mean difference is $2\Delta_1$, the relative entropy $D(P_1 \| P_2)$ of

the observed-transcript laws under ADV_1 and ADV_2, respectively, is bounded by a constant times $n\Delta_1^2/K = n/H_{\mathrm{UNIF}}$, where $H_{\mathrm{UNIF}} = K/\Delta_1^2$ is the reference complexity. The change-of-measure and testing bound in Lemma 10 gives

$$P_1(J_n \neq 1) + P_2(J_n = 1) \geq \frac{1}{2}\exp\left(-\frac{32n}{H_{\mathrm{UNIF}}}\right).$$

There are different intended best arms, but too few informative observations to distinguish the problems reliably.

Concentration of the complete reward tables preserves the intended best arms, 1 and $\overline{k}$, and membership in $\boldsymbol{\Delta}_3$, except on sufficiently rare tables. Removing these bad tables from the testing bound and averaging over the remaining finite supports yields two deterministic witnesses in $\boldsymbol{\Delta}_3$, with the required error rate on at least one of them. The appendix gives the complete argument.

Remark 3 *The condition $K\exp(-n\Delta_1^2/128) \leq 1/128$ controls fluctuations of the complete reward tables. Together with $K \geq 4096$, it makes the bad-table probabilities small relative to the exponentially small testing lower bound. The assumption on K is likely to be an artifact of the proof. The lower bound is a worst-case statement over the prescribed class, at the exponential scale n/H_{UNIF}. The comparison with uniform sampling should be understood at this scale; it does not rule out better performance on particular instances.*

4. The best of both worlds challenge

In this section, we ask if we can have a learner that performs optimally under adversarial and stochastic rewards. The lower bound in Theorem 4 shows that in general, this is impossible.

Existing robust solutions? In the stochastic setting, a state-of-the-art algorithm, Sequential Halving (`SH`, Karnin et al., 2013)—see also Successive Rejects (`SR`) by Audibert et al. (2010)—guarantees $e_{\mathrm{STO}}(n) \leq \mathcal{O}\left(\log K \exp\left(-n/(H_{\mathtt{SR}}\log K)\right)\right)$. However, as discussed in the introduction, `SR` or `SH` can fail against a worst-case adversary. On the other hand, as discussed by Audibert et al. (2010), uniform-like algorithms (like `Rule`) can only guarantee that in the stochastic case, we get $e_{\mathrm{STO}}(n) \leq \widetilde{\mathcal{O}}\left(\exp\left(-n/H_{\mathrm{UNIF}}\right)\right)$. In general, $H_{\mathtt{SR}} \leq H_{\mathrm{UNIF}}$ and in some problems, we even have $H_{\mathtt{SR}} = (2/K)H_{\mathrm{UNIF}}$. Therefore, `SH` can notably outperform uniform algorithms in the stochastic case.

The best of both worlds This leads to the central question of the paper: Can a learner achieve optimal error rates in both settings without knowing whether the rewards are stochastic or adversarial? More precisely, the question is whether such a learner can simultaneously guarantee, for every horizon n, every stochastic problem STO, and every adversarial reward sequence $\boldsymbol{g}$, the following bounds on its respective probabilities of misidentification $e_{\mathrm{STO}}(n)$ and $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$:

$$e_{\mathrm{STO}}(n) \leq \widetilde{\mathcal{O}}\left(\exp\left(-\frac{n}{H_{\mathtt{SR}}\log K}\right)\right) \quad \text{and} \quad e_{\mathrm{ADV}(\boldsymbol{g})}(n) \leq \widetilde{\mathcal{O}}\left(\exp\left(-\frac{n}{H_{\mathrm{UNIF}}(\boldsymbol{g})}\right)\right).$$

Why is the BOB question challenging? The learner could choose, for arm k, at round t after it has been sampled at least once, to use the cumulative gain estimator $\widehat{G}_{k,t} = \frac{t\sum_{t'=1}^{t}\mathbf{1}\{I_{t'}=k\}g_{k,t'}}{\sum_{t'=1}^{t}\mathbf{1}\{I_{t'}=k\}}$. This estimator can be potentially highly biased if it is used against a malignant adversary. For

this reason, we base our approach on the estimator $\widetilde{G}_{k,t} = \sum_{t'=1}^{t} \frac{\boldsymbol{g}_{k,t'}}{\boldsymbol{p}_{k,t'}} \mathbf{1}\{I_{t'} = k\}$. However, this usage potentially introduces high variance in our estimates; the cumulative conditional variance of $\widetilde{G}_{k,t}$ relevant to martingale concentration, namely the sum of the one-step conditional variances, is bounded by $\sum_{t'=1}^{t} 1/\boldsymbol{p}_{k,t'}$. The high variance is most damaging in the stochastic case when trying to have a learner based on $\widetilde{G}_{k,t}$ to obtain the optimal error rates of Karnin et al. (2013). Indeed, these optimal rates are obtained by algorithms using empirical means, as in $\widehat{G}_{k,t}$, for which strong concentration guarantees are available in the stochastic setting. Therefore, we strive to characterize the minimum amount of unavoidable variance of the mean estimators of each arm. The learner would like to allocate more pulls at any round t to the arms that are among the best arms, which means having large $\boldsymbol{p}_{k,t}$ for these arms. Indeed, discriminating between them is the hardest part of the task and large $\boldsymbol{p}_{k,t}$ reduces the variance term $1/\boldsymbol{p}_{k,t}$. However, it is natural to think that the learner is not able to guarantee that it pulls the best arms at the beginning of the game more than in a uniform fashion. If the arms are pulled uniformly, each one-step conditional variance is at most K, which can be large. The time spent accumulating variance before the best arms are well identified influences the concentration bound and hence the bound on the final probability of error. Intuitively, the lower bound in Theorem 4 constructs worst-case examples showing that any learner cannot pull the best arm more than a certain amount in some period of the game because it is difficult to identify the best arms. Therefore, this learner is susceptible to be tricked by an adversary. Our new learner in Section 5 tries to limit this effect by making early and almost costless bets on what are the best arms given the early rewards and starts to pull them more right away. Note that a long initial uniform sampling phase can contribute a term of order nK to the conditional-variance upper bound for the importance-weighted cumulative estimator $\widetilde{G}_{k,n}$. This can limit the concentration guarantee obtained from that bound.

Another approach could be a learner that determines online if the observed rewards are stochastic. This was used by Bubeck and Slivkins (2012) and Auer and Chiang (2016). They detect if the difference between $\widehat{G}_{k,t}$ and $\widetilde{G}_{k,t}$ is way too large. However, their bound itself uses terms depending on K and the variance of each arm, $1/\boldsymbol{p}_{k,t}$, which leads to similar open questions as discussed just above. In this paper, inspired by the approach of Seldin and Slivkins (2014), we give a practical simple parameter-free and versatile algorithm. Furthermore, the algorithms that are based on stochastic tests are usually cumbersome and complex, as discussed by Seldin and Slivkins, 2014.

Why is the best of both worlds unachievable? We define a new notion of complexity, H_{BOB} as

$$H_{\mathrm{BOB}} \triangleq \frac{1}{\Delta_{(1)}} \max_{k \in [K]} \frac{k}{\Delta_{(k)}}.$$

The complexity H_{BOB} captures the difficulty of stochastic best-arm identification when the learner must also remain robust to adversarial rewards. As we detail in Remark 5, $H_{\mathtt{SR}} \leq H_{\mathrm{BOB}} \leq H_{\mathrm{UNIF}}$. The gap between $H_{\mathtt{SR}}$ and H_{BOB} reflects the additional difficulty imposed by adversarial robustness.

Theorem 4 (Lower bound for the BOB challenge) *Fix $K \geq 2$ and reference gaps $0 < \Delta_1 = \Delta_2 \leq \cdots \leq \Delta_K \leq 1/8$, and let $\boldsymbol{\Delta}_4$ be the corresponding problem class. For any learner and any positive integer horizon n satisfying*

$$K \exp\left(-\frac{\Delta_1^2 n}{128}\right) \leq \frac{1}{32},$$

there exists an i.i.d. stochastic problem $\mathrm{STO} \in \boldsymbol{\Delta}_4$ *such that, if*

$$e_{\mathrm{STO}}(n) \leq \frac{1}{64} \exp\left(-\frac{2048n}{H^{\mathrm{STO}}_{\mathrm{BOB}}}\right),$$

then there exists a deterministic gain table $\boldsymbol{g} \in [0,1]^{K\times n}$ *belonging to* $\boldsymbol{\Delta}_4$ *for which*

$$e_{\mathrm{ADV}(\boldsymbol{g})}(n) \geq \frac{1}{16}.$$

Here $H^{\mathrm{STO}}_{\mathrm{BOB}}$ *is the complexity computed from the gaps of the constructed stochastic problem. The constructed stochastic problem and gain table may depend on the learner and on* n*.*

Remark 5 *In general, we have*

$$H_{\mathtt{SR}} \leq H_{\mathrm{BOB}} \leq H_{\mathrm{UNIF}}.$$

Below, we compare the three complexities in three specific gap regimes in order to intuitively explore whether the inequalities in the previous equation are strict or not. Interestingly, while in two regimes $H_{\mathtt{SR}} = H_{\mathrm{BOB}}$*, in the third regime, called the 'square-root gaps', we can obtain* $H_{\mathrm{BOB}} = \sqrt{K/2}H_{\mathtt{SR}}$*. This equality shows that on some problems and for large values of* K*, our lower bound on the complexity of the BOB problem is significantly larger than the complexity of the strictly stochastic case. Uniformly scaling the gaps down to satisfy* $\Delta_{(K)} \leq 1/8$ *leaves all complexity ratios unchanged. The square-root separation also persists, up to universal constants, for the stochastic instances constructed in the proof of Theorem 4 (Appendix E). For sufficiently large* K*, the optimal purely stochastic rate, if guaranteed uniformly over stochastic instances and horizons, would satisfy the premise of Theorem 4 at all sufficiently large horizons. Hence it cannot be combined with vanishing worst-case error over the adversarial class. Thus, in general, the best performance achievable in the two worlds cannot be obtained simultaneously.*

▶ **Flat regime** We assume all the gaps are equal, $k \in [K],\ \Delta_k = \Delta_1$. Then, $H_{\mathtt{SR}} = H_{\mathrm{BOB}} = H_{\mathrm{UNIF}}$. Having $H_{\mathtt{SR}} = H_{\mathrm{BOB}}$ shows that our stochastic BOB lower bound for robust learners (Theorem 4) using H_{BOB} is of the same order as the one in the strict stochastic setting (Audibert et al., 2010) using $H_{\mathtt{SR}}$. In this stochastic regime, `Rule` is optimal while being robust to an adversary.

▶ **Super-linear gaps** Let $2 \in \operatorname{argmin}_{k\in[K]}(\Delta_{(k)}/k)$. This holds if $\Delta_{(k)} = k\Delta_{(1)}$ for every rank k with $3 \leq k \leq K$, and $\Delta_{(1)} \leq 1/(8K)$. Then, $H_{\mathtt{SR}} = H_{\mathrm{BOB}} = (2/K)H_{\mathrm{UNIF}}$. Again, our BOB lower bound is of the same order as in the strict stochastic setting. This seems to indicate that in this case, BOB is achievable. However, it is not achieved by the uniform learner that is clearly suboptimal. This motivates the design of a learner that adapts its sampling strategy. Intuitively, the learner can identify bad arms quickly and start focusing early on the best arms without incurring high variance on its estimates for them.

▶ **Square-root gaps** We assume $2 \in \operatorname{argmin}_{k\in[K]}(\Delta^2_{(k)}/k)$.
Choose a rank $j \in \operatorname{argmin}_{k\in[K]}(\Delta_{(k)}/k)$. For some constant c, let $\Delta_{(1)} = c\Delta_{(j)}/j$. We have $c \leq \sqrt{2j}$ because $\Delta^2_{(1)}/2 \leq \Delta^2_{(j)}/j$ as $2 \in \operatorname{argmin}_{k\in[K]}(\Delta^2_{(k)}/k)$. Therefore,

$$H_{\mathtt{SR}} = \frac{2}{\Delta^2_{(2)}} = \frac{2}{\Delta^2_{(1)}} = \frac{2}{\Delta_{(1)}}\frac{j}{c\Delta_{(j)}} = \frac{2H_{\mathrm{BOB}}}{c} = \frac{2H_{\mathrm{UNIF}}}{K}.$$

We can get $c = \sqrt{2j} = \sqrt{2K}$. This happens if $\sqrt{K/2}\Delta_{(1)} = \Delta_{(K)}$ and $\sqrt{k/2}\Delta_{(1)} \geq \Delta_{(k)}$ for ranks k with $3 \leq k < K$. Then, we get $\sqrt{K/2}H_{\mathrm{SR}} = H_{\mathrm{BOB}}$. H_{BOB} is $\sqrt{K/2}$ larger than the complexity of the strictly stochastic setting. Intuitively, the learner needs to spend some time to identify the 'square-root gaps' suboptimal arms before starting to focus on the best arms. This makes it suffer an additional amount of variance on its estimates for the best arms.

Sketch of the proof *(full proof in Appendix E)*: The proof constructs a stochastic problem and an adversarial environment that are difficult for the learner to distinguish, but for which successful identification requires different recommendations. Our proof of the lower bound uses some arguments of purely stochastic best-arm identification lower bounds of Audibert et al. (2010) and Carpentier and Locatelli (2016). We were also inspired by the lower bound of Auer and Chiang (2016) for the BOB question for the cumulative regret. However, our specific construction is new.

Fix the reference gaps, a learner, and a horizon n satisfying Theorem 4. We construct a stochastic and an adversarial problem. Between the two problems, only one arm differs. Let us define K base Bernoulli distributions with means $\mu_1^{\mathrm{BASE}} = 1/2$ and $\mu_j^{\mathrm{BASE}} = 1/2 - \Delta_j$ for $j \in [2 : K]$. Choose any maximizing index $i \in \arg\max_{j\in[K]} j/\Delta_j$; since $\Delta_1 = \Delta_2$, $i \geq 2$. The initial phase is the prefix of n_i rounds, where

$$a_i = \min\left\{1, \frac{2\Delta_1}{\Delta_i}\right\}, \qquad n_i = \lceil na_i \rceil \leq n.$$

If $n_i = n$, as when $i = 2$, the second phase is empty.

By Dirichlet's box principle, for $i \geq 3$ at least one arm, denoted $\overline{k}$, among $\lceil i/2 \rceil, \ldots, i$ has expected pull count at most $2n_i/i$. The maximizing property of i makes its gap comparable to Δ_i. For $i = 2$, take $\overline{k} = 2$. In both cases,

$$\mathbb{E}_{\mathrm{BASE}}[T_{\overline{k}}(n_i)] \leq \frac{2n_i}{i}, \qquad \frac{\Delta_i}{2} \leq \Delta_{\overline{k}} \leq \Delta_i, \qquad \overline{k} \neq 1.$$

This lightly sampled arm is chosen from expectations under BASE, before sampling the constructed tables.

The stochastic problem STO is made by only modifying the original Bernoulli distribution of arm $\overline{k}$, setting its mean to $1/2 + \Delta_1/4$. Its distributions are stationary, and $\overline{k}$ is uniquely best. The adversarial problem ADV samples $\boldsymbol{g}$ randomly before interaction, independently of the learner's randomness. It follows BASE through the first n_i rounds and STO afterward; all other arms keep their distributions throughout. To make the resulting lower bound as strong as possible, we seek a short initial phase to limit the distinguishing information. However, it must be long enough to keep arm 1 uniquely best on a high-probability good-table event and preserve the prescribed gaps after reward fluctuations. The chosen fraction provides this margin while remaining of order Δ_1/Δ_i, up to the cap at one. The appendix verifies that STO and every good ADV table belong to $\boldsymbol{\Delta}_4$.

Let P_A, P_S be the distributions of the learner's observed actions, rewards, and final recommendation under ADV and STO. To compare these observations, we use the change-of-measure method of Kaufmann, Cappé, and Garivier (2016), in the form proved in Lemma 10. The lemma expresses the relative entropy of the two observation distributions as the expected number of informative pulls times the information supplied by each pull:

$$D(P_A \| P_S) = \mathbb{E}_{\mathrm{BASE}}[T_{\overline{k}}(n_i)]\,\mathrm{kl}\left(\frac{1}{2} - \Delta_{\overline{k}}, \frac{1}{2} + \frac{\Delta_1}{4}\right).$$

Here the ADV and BASE interaction laws agree through the prefix. The appendix gives, for universal constants C, C',

$$D(P_A \| P_S) \leq \frac{C n_i \Delta_i^2}{i} \leq \frac{C' n \Delta_1 \Delta_i}{i} = \frac{C' n}{H_{\mathrm{BOB}}^{\mathrm{BASE}}}.$$

We want the lower bound on the learner's error to be as large as possible. When the learner meets the prescribed guarantee on STO, a smaller upper bound on $D(P_A \| P_S)$ forces a larger lower bound on its error in ADV: its observations provide less evidence for distinguishing the two problems. This explains our choice of i: maximizing i/Δ_i minimizes the expression $n\Delta_1\Delta_i/i$ in the bound above. For this maximizing index, $H_{\mathrm{BOB}}^{\mathrm{BASE}} = i/(\Delta_1\Delta_i)$, which gives the dependence on $n/H_{\mathrm{BOB}}^{\mathrm{BASE}}$. The Δ_j are the reference gaps; after reordering the constructed STO gaps, the appendix shows $H_{\mathrm{BOB}}^{\mathrm{STO}} \leq 16 H_{\mathrm{BOB}}^{\mathrm{BASE}}$. Thus the information is also bounded by a universal constant times $n/H_{\mathrm{BOB}}^{\mathrm{STO}}$.

Recommending arm 1 is an error in STO. If its error satisfies Theorem 4's premise, this recommendation is very unlikely under P_S. The lemma's event consequence prevents it from being sufficiently likely under P_A for reliable identification of arm 1, forcing a positive constant adversarial error. Concentration controls the bad tables, and the appendix extracts a deterministic $\boldsymbol{g} \in \boldsymbol{\Delta}_4$ with error at least $1/16$. The constructed instances may depend on n; this suffices against guarantees required uniformly over instances and budgets.

5. A simple robust parameter-free algorithm for stochastic & adversarial rewards

In this section, we present a new learner and analyze its theoretical performance against any i.i.d. stochastic problem or any adversarially designed rewards.

We call the algorithm `ProbabilityONE`, denote it by `P1`, and detail it in Figure 2. Intuitively, `P1` pulls the estimated best arm with "probability" one, the estimated second best arm with "probability" one half, the estimated third best arm with "probability" one third, and so on until pulling the estimated worst arm with "probability" one over K.[a] In order to get proper probabilities, we need to normalize them by the K-th harmonic number $\overline{\log} K = \sum_{k=1}^{K}(1/k)$, where $\overline{\log} K \leq \log K + 1$ for all positive integers K. `P1` uses a Zipf distribution over the estimated ranks with an exponent of 1 (Powers, 1998).

For $t = 1, 2, \ldots$

- Sort and rank the arms by decreasing order of $\widetilde{G}_{\cdot,t-1}$: Rank arm k as $\widetilde{\langle k \rangle}_t \in [K]$[a].
- Select arm $I_t \in [K]$ with $\boldsymbol{p}_{k,t} \triangleq \mathbb{P}\left(I_t = k \mid \mathcal{F}_{t-1}\right) \triangleq \dfrac{1}{\widetilde{\langle k \rangle}_t \, \overline{\log} K}$ for all $k \in [K]$.

Here $\mathcal{F}_{t-1}$ is the information available before sampling, including the resolved ranking but excluding the fresh action draw and current rewards.

Recommend, at any given round t, $J_t \triangleq \operatorname{argmax}_{k \in [K]} \widetilde{G}_{k,t}$.

a. Equalities between arms or comparisons with arms that have not been pulled yet are broken arbitrarily.

Figure 2: The `ProbabilityONE` (`P1`) algorithm

The estimate used in P1 is $\widetilde{G}_{k,t-1}$ for arm k at round t. P1 is heavily inspired by Successive Rejects (SR) of Audibert et al. (2010), as both rank the arms and attempt to allocate to arm k a number of pulls $\widetilde{\mathcal{O}}\left(n/\langle k\rangle\right)$ according to its true rank $\langle k\rangle$. Our new learner is parameter-free and naturally anytime (agnostic of n). Like SR, P1 requires neither prior knowledge nor estimation of the complexity parameters. Unlike SR, P1 does not divide the game into different sampling phases. P1 follows the same deliberately simple rank-based sampling rule at every round.

As discussed in Section 4, in order to minimize the misidentification error in the stochastic case, it is crucial to limit the variance of the estimators for the best arms. Therefore P1, from its very first pull, pulls more (i.e., with higher probability) the arms that are estimated to be among the best. First, this comes with almost no cost: Indeed, pulling the estimated best arm with probability $1/\overline{\log}K$ does not prevent from pulling all the arms almost uniformly and more precisely with probability at least $1/(K\overline{\log}K)$. Therefore, no suboptimal arm is actually left out in the early chase for the best arm, and the bounds $1/\boldsymbol{p}_{k,t}$ on the conditional variances are at most $\overline{\log}K$ times the corresponding bound under uniform sampling. Second, it gives the learner the flexibility to adapt to different types of stochastic problems with different gap regimes. If in a setup, some arms are clearly suboptimal, it is helpful to pull the clearly best arm more and right from the beginning. This is a more flexible behavior than the one of algorithms that are fixing the number of pulls for each arm during a fixed period in advance.

The key idea is that when P1 ranks a promising arm highly, it samples that arm more frequently, reducing the variance of its importance-weighted estimates. This, in turn, helps improve subsequent rankings. We capture this feedback through virtual comparison times that are introduced only for the analysis and are not used by the algorithm.

Additionally, compared to a fixed-phase algorithm, our analysis is also more flexible: We can analyze, for instance, the quality of the estimated ranking $\widetilde{\langle\cdot\rangle}$ and therefore the adaptive sampling procedure of the arms at any round. Actually, these rounds called *comparison rounds*, can be chosen in a problem-dependent manner, in order to minimize the final probability of error. This is reflected in the complexity measure present in the upper bound, which is defined as the minimum over admissible choices of comparison times. P1 remains agnostic to the optimal 'virtual' phases, which are introduced only for the analysis. We now define this new notion of complexity. First, we define the proportion of rounds for comparison $\boldsymbol{a}$ in a space $\boldsymbol{A}$. Let $\boldsymbol{A} \triangleq \left\{\boldsymbol{a} \in [0,1]^K : na_i \in \mathbb{N}, \forall i \in [K], 1 = a_1 = a_2 \geq a_3 \geq \ldots \geq a_K > a_{K+1} = 0\right\}$. The complexity associated with the P1 learner is $H_{\texttt{P1}}$ and is defined first as

$$H_{\texttt{P1}}(\boldsymbol{a}) \triangleq \max_{k\in[K]} \frac{\sum_{i=\langle k\rangle}^{K}(a_i - a_{i+1})i + \frac{1}{24}Ka_{\langle k\rangle}\Delta_k}{a_{\langle k\rangle}^2\Delta_k^2}\overline{\log}K \quad \text{and then,} \quad H_{\texttt{P1}} \triangleq \min_{\boldsymbol{a}\in\boldsymbol{A}} H_{\texttt{P1}}(\boldsymbol{a}).$$

In $H_{\texttt{P1}}(\boldsymbol{a})$, for arm k, the term $\sum_{i=\langle k\rangle}^{K}(a_i - a_{i+1})i$ corresponds to the sum of variances $\widetilde{\mathcal{O}}(i)$, for $i \in [\langle k\rangle : K]$, during a proportion of time $a_i - a_{i+1}$ between the comparison rounds $i+1$ and i. Indeed, on the good-ranking event described below, for $i < K$ and any arm k satisfying $\mu_{(1)} - \mu_k < \Delta_{(i+1)}/2$, we have $\widetilde{\langle k\rangle}_t \leq i$ during $na_{i+1} < t \leq na_i$. For $i = K$, the same rank bound is automatic. This leads, on these events and at the corresponding sampling rounds, to $\boldsymbol{p}_{k,t} \geq 1/(i\overline{\log}K)$ and therefore, as $\widetilde{\boldsymbol{g}}_{k,t} = \boldsymbol{g}_{k,t}\mathbf{1}\{I_t = k\}/\boldsymbol{p}_{k,t}$, to a conditional variance of $\widetilde{\boldsymbol{g}}_{k,t}$ at most $i\overline{\log}K$. In the denominator, $a_{\langle k\rangle}$ specifies the comparison time $na_{\langle k\rangle}$, the number of elapsed rounds available before that comparison. The additional term involving $K/24$ comes from the linear correction in the Bernstein–Freedman concentration inequality.

Theorem 6 (Upper bounds for P1) *For any stochastic problem* STO *with complexity $H_{\mathtt{P1}}$ and for any $\boldsymbol{g}$ fixed by an oblivious adversary with complexity $H_{\mathrm{UNIF}}(\boldsymbol{g})$, the probabilities of error of* P1*, denoted $e_{\mathrm{STO}}(n)$ and $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$ in their respective environment, for any n, simultaneously verify*

$$e_{\mathrm{STO}}(n) \leq 2K^3 n \exp\left(-\frac{n}{128 H_{\mathtt{P1}}}\right) \quad \text{and} \quad e_{\mathrm{ADV}(\boldsymbol{g})}(n) \leq K \exp\left(-\frac{3n}{40\overline{\log}(K) H_{\mathrm{UNIF}}(\boldsymbol{g})}\right).$$

Sketch of the proof *(full proof in Appendix F)*: For the *adversarial case*, it is enough that the learner pulls each arm with probability at least $1/(K\overline{\log}K)$ to obtain the same complexity H_{UNIF} as Rule, up to a factor $\log K$. For the *stochastic case*, we consider $K-1$ arbitrary 'virtual' phases that each ends at round $n_i = na_i$. These comparison times are fixed for the analysis and optimized only after deriving the final error bound. Note that P1 is oblivious to these values. The phases are indexed in reverse order, from K down to 2. Intuitively, n_i is a round after which we expect the following event ξ_i to happen with high probability: For all integer u with $n_i < u \leq n+1$, the ranking from estimates through $u-1$ satisfies $\widetilde{\langle k \rangle}_u \leq i-1 < i$ whenever $\mu_{(1)} - \mu_k < \Delta_{(i)}/2$. At $u = n+1$, this is the virtual final ranking consistent with the recommendation. The important consequence is that, on this event, any such arm k is pulled with $\boldsymbol{p}_{k,t} \geq 1/((i-1)\overline{\log}K)$ at sampling rounds $n_i < t \leq n$, giving conditional variance at most $(i-1)\overline{\log}K$ in its estimates $\widetilde{\boldsymbol{g}}_{k,t}$. Reducing these variances leads in turn to better estimates in the rest of the game. The proof works iteratively over the phases. We bound the probability that ξ_i fails together with the required earlier ranking guarantees. On these good-ranking events, the conditional variance is controlled, so martingale concentration bounds their joint occurrence with a later comparison error. Summing these joint error probabilities gives a bound on the probability of an incorrect final recommendation, based on $n_2 = n$ rounds of observations.

To bound $e_{\mathrm{STO}}(n)$, we use the Bernstein inequality for martingale differences. This inequality takes into account the variances and holds despite the dependencies of the random variables $\widetilde{\boldsymbol{g}}_{k,t}$.

When minimizing $H_{\mathtt{P1}}$ by choosing a_k, for $k \in [K]$, we need to trade off between short phases, possibly meaning not enough samples to discriminate the suboptimal arms (the denominator term of the suboptimal arms is small) and long phases, which means that in early stages, the best arms are considered as badly ranked for a long time and the variance of their mean estimators is increasing with the length of the early phases (the numerator term for the best arms is larger).

Corollary 7 *For $n \geq \Delta_{(K)}/\Delta_{(1)}$, the complexity $H_{\mathtt{P1}}$ of* P1 *matches the complexity H_{BOB} from the lower bound of Theorem 4 up to log factors; more precisely,*

$$H_{\mathtt{P1}} \leq 2H_{\mathrm{BOB}}(\overline{\log}K)^2 = \mathcal{O}\left(H_{\mathrm{BOB}} \log^2 K\right).$$

For every positive integer horizon n, the stochastic probability of error of P1 *satisfies*

$$e_{\mathrm{STO}}(n) \leq 2K^3 n \exp\left(-\frac{n}{256 H_{\mathrm{BOB}}(\overline{\log}K)^2}\right).$$

The result of Corollary 7 is obtained by setting $a_k = \left\lceil n\Delta_{(1)}/\Delta_{(k)} \right\rceil / n$, $\forall k \in [K]$. We round each ideal comparison time upward to the next round. This changes only the analysis, leaving the learner and its horizon unchanged, and costs at most a factor two in the complexity in the stated budget range. For smaller budgets, the displayed probability bound is trivial. Notice that the same gap ratio also guides the lower-bound construction of Theorem 4. For its reference gaps, we use

$a_i = \min\{1, 2\Delta_1/\Delta_i\}$ to leave enough margin in the adversarial gaps after allowing for random fluctuations. The full proof of Corollary 7 is in Appendix G, where we also discuss the relation between $H_{\texttt{P1}}$ and $H_{\texttt{SR}}$ and $H_{\texttt{BOB}}$ for different regimes of the gaps. Corollary 7 demonstrates that P1 achieves the best that can be wished for in the two worlds, up to log factors.

Remark 8 *In the adversarial case, a modification to* P1 *could lead to a similar upper bound as for* Rule*, where* $H_{\text{UNIF}}(\boldsymbol{g})$ *appears instead of* $H_{\text{UNIF}}(\boldsymbol{g})\overline{\log} K$*. At each round, with probability one half we can play according to* Rule *and otherwise use* P1*. The importance-weighted estimator uses the actual sampling probability* $\boldsymbol{p}_{k,t}$ *of this mixture. We keep the recommendation* $J_n = \operatorname{argmax}_{k\in[K]} \widetilde{G}_{k,n}$*. Such a mixture could retain useful stochastic adaptation.*

Remark 9 *We studied the hard (adversarial) and easy data (stochastic) settings. However, as discussed by Seldin and Slivkins (2014) and Allesiardo et al. (2017), we can consider intermediate settings of difficulty. First, one could investigate whether the result in the stochastic case holds up to constants when the gaps of the arms do not change by more than the same numerical factor during the game. More generally, we could design variants of* $H_{\texttt{P1}}$ *under the assumption that after some round* t' *the (ground-truth) ranks of all the arms are upper bounded each by a constant. A variant of* P1 *might adapt after* t' *and rank every arm at most according to its maximum rank. Such results might even hold in a case of a change of the identity of the best arm in the game.*

6. Possible extensions of P1

Fixed-confidence In the fixed-confidence setting for i.i.d. stochastic rewards (Maron and Moore, 1993, Even-Dar et al., 2006; Mnih et al., 2008; Kalyanakrishnan et al., 2012; Kaufmann and Kalyanakrishnan, 2013; Garivier and Kaufmann, 2016), the goal is to design a learner that stops as soon as possible and returns the best arm with a fixed confidence. Let $\widetilde{n}$ be the round when the algorithm stops and $J_{\widetilde{n}}$ its returned arm. Given a confidence level δ, the learner has to guarantee that $\mathbb{P}\left(J_{\widetilde{n}} \neq k^\star_{\text{STO}}\right) \leq \delta$. The performance of the learner is then measured by its *sample complexity*, which is the number of rounds $\widetilde{n}$ before stopping, either in expectation or in high probability.

Mimicking the Hoeffding and Bernstein Races (Maron and Moore, 1993; Mnih et al., 2008), we could design Freedman Races based on P1. All the arms would be pulled according to the parameter-free sampling routine of P1. No arm could ever be discarded because we could happen to face an adversary. The learner would stop using the P1 routine based on having the confidence intervals from the Bernstein concentration inequality for martingales and in particular, it would stop when the confidence interval for the empirically best arm is separated from the confidence intervals for all the other arms. For this fixed-confidence variant of P1, we could reuse the proof techniques developed for the fixed-budget setting and bound the accumulation of variance of the estimates. Then, in the stochastic case, we conjecture that the expected sample complexity of such algorithm is $\widetilde{\mathcal{O}}\left(H_{\text{BOB}} \log(1/\delta)\right)$, up to log factors.

For the adversarial case, we can consider an infinite sequence of rewards $\boldsymbol{g}$ fixed by the adversary for all arms. Assume that the sample complexity on the rewards up to round t, is bounded by some $\boldsymbol{n}(\boldsymbol{g}_{[t]})$, the smaller the better. A natural target would be that if at any round t, $\boldsymbol{g}_{[t]}$ verifies $\boldsymbol{n}(\boldsymbol{g}_{[t]}) \leq t$, then with probability $1-\delta$, the learner can both stop and recommend the best arm $k^\star_{\boldsymbol{g}_{[t]}}$ at round t. We conjecture that our Freedman algorithm could satisfy this requirement with the complexity of uniform allocation, $\widetilde{\mathcal{O}}\left(H_{\text{UNIF}}(\boldsymbol{g}_{[t]}) \log(1/\delta)\right)$. Note that the learner could possibly never stop.

Streams, windows, thresholds, m-set, and active anomaly detection P1 could be adapted to recommend the best arm in the latest window of W rounds ending at $t \geq W$. One could investigate bounds with W replacing n when a variant of P1 recommends the estimated best arm in that window. P1 could also be adapted to identify m best arms out of K as did Bubeck et al. (2013). A key would be to redefine the gaps around the boundary of the set of m best arms. We could also extend P1 to a setting where the rewards converge (Li et al., 2016). Locatelli et al. (2016) defined the gaps with respect to a given fixed threshold and the objective is to determine which arms have a mean higher than the threshold. Again, our approach could apply. Moreover, it could prove to be a good robust approach to the problems that are linked to the threshold bandit problem as discussed by Locatelli et al. (2016, Section 3), one of them being the *active anomaly detection* (Carpentier and Valko, 2014). Indeed, in adversarial anomaly detection, the learner might be monitoring different streams of non-stochastic rewards and could potentially detect an anomaly if one of the streams outputs a reward signal that is on average larger than a given threshold during a time window period W. A variant of P1 could then be a robust anomaly detector and could be also adaptive to easy data.

Adaptive adversaries Our upper bounds might extend naturally to adaptive adversaries given the following condition: H_{UNIF} is an upper bound on the complexity of all $\boldsymbol{g}$ that the adaptive adversary can possibly generate. A possible route would be to replace the Bernstein concentration inequality in Theorem 1 by the Bernstein concentration inequality for martingales.

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Appendix A. Experiments in the stochastic setting

We complement our theoretical results with experiments in the stochastic setting. We reuse the experimental setups of Audibert et al. (2010) in Experiments 1 to 7. We only consider Bernoulli distributions and the optimal arm always has mean $1/2$. Each experiment corresponds to a different situation for the gaps. They are either clustered in a few groups or distributed according to an arithmetic or geometric progression. Experiment 8 reuses the 'square-root gap' scenario when $H_{\text{BOB}} = \sqrt{K/2}H_{\text{SR}}$ as detailed in Remark 5. The experimental setups are given below.

- Experiment 1: *One group of bad arms*, $K = 20$, $\mu_{2:20} = 0.4 \equiv \forall j \in \{2, \dots, 20\}, \mu_j = 0.4$
- Experiment 2: *Two groups of bad arms*, $K = 20$, $\mu_{2:6} = 0.42$, $\mu_{7:20} = 0.38$.
- Experiment 3: *Geometric progression*, $K = 4$, $\mu_i = 0.5 - (0.37)^i$, $i \in \{2, 3, 4\}$
- Experiment 4: *6 arms divided into three groups*, $K = 6$, $\mu_2 = 0.42$, $\mu_{3:4} = 0.4$, $\mu_{5:6} = 0.35$
- Experiment 5: *Arithmetic progression*, $K = 15$, $\mu_i = 0.5 - 0.025i$, $i \in \{2, \dots, 15\}$
- Experiment 6: *2 good arms and a large group of bad arms*, $K = 20$, $\mu_2 = 0.48$, $\mu_{3:20} = 0.37$
- Experiment 7: *Three groups of bad arms*, $K = 30$, $\mu_{2:6} = 0.45$, $\mu_{7:20} = 0.43$, $\mu_{21:30} = 0.38$
- Experiment 8: *Square-root gaps* $K = 100$, $\mu_i = 0.5 - 0.25\sqrt{i/(2K)}$, $i \in [2 : 100]$

In Table 1, we report the complexities H_{SR}, H_{BOB}, and H_{UNIF} computed in these experimental setups. Unsurprisingly, in Experiments 1, 3, and 5 we recover $H_{\text{SR}} = H_{\text{BOB}}$ and in Experiment 8, we have $H_{\text{BOB}} = \sqrt{K/2}H_{\text{SR}}$. Experiments 2, 4, 6, and 7 then give an idea about the behavior of H_{SR}, H_{BOB}, and H_{UNIF} with respect to each other.

Experimental setup	H_{SR}	H_{BOB}	H_{UNIF}
1. One group of bad arms	2000	2000	2000
2. Two groups of bad arms	1389	2083	3125
3. Geometric progression	5540	5540	11080
4. 6 arms divided into three groups	400	500	938
5. Arithmetic progression	3200	3200	24000
6. 2 good arms and a large group of bad arms	5000	7692	50000
7. Three groups of bad arms	4082	5714	12000
8. Square-root gaps	3200	22627	160000

Table 1: Comparing complexities H_{SR}, H_{BOB}, and H_{UNIF}.

Figure 3 reports the empirical error rates of `SR`, `P1`, and *static uniform allocation* across the eight experimental settings. The static uniform allocation is not the algorithm `Rule`. `Rule` samples an arm uniformly at random while the *static* uniform allocation simply allocates n/K pulls to each arm deterministically. The empirical results follow very closely our theoretical findings as the empirical behavior in Figure 3 mimics the behavior of the complexities in Table 1. Following Audibert et al. (2010), we chose the horizon n to be of the order of the complexity $H_1 = \sum_{k \in [K]}(1/\Delta_k^2)$, where $H_{\text{SR}} \leq H_1 \leq H_{\text{SR}}\overline{\log}K$.

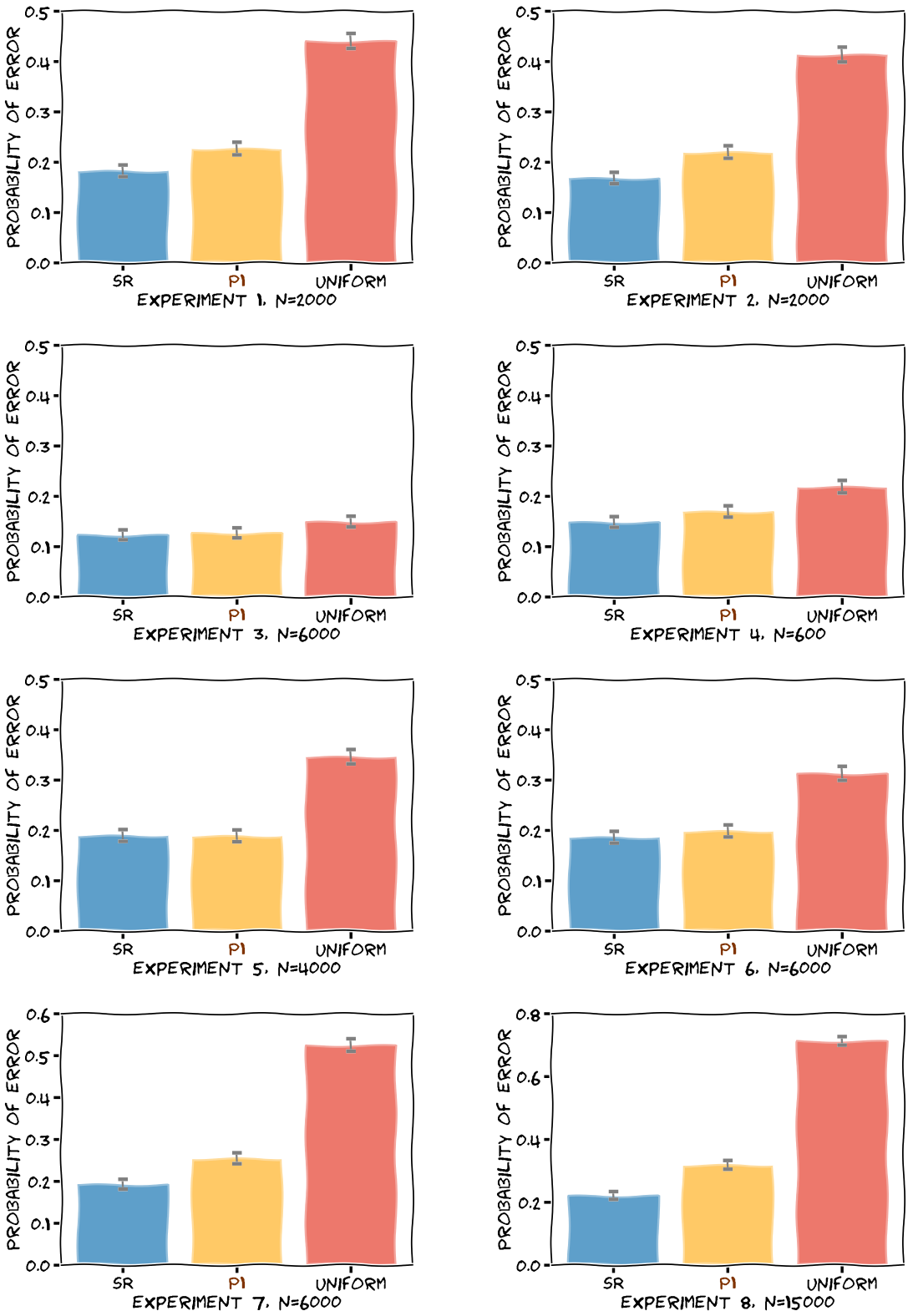


Figure 3: Probabilities of error of SR, P1, and the static uniform allocation.

In the remainder of the appendix, for a random variable X, we denote its variance by σ_X^2. Moreover, we also write that a bounded random variable $X \in [\ell_X, u_X]$ has a range $b_X = u_X - \ell_X$.

Appendix B. Upper bound on the probability of error of `Rule` in the general adversarial case

Theorem 1 (Upper bound for `Rule` in the adversarial case) *For any horizon n, given rewards $\boldsymbol{g}$ chosen by an oblivious adversary, with $\boldsymbol{g}_{k,t} \in [0,1]$ for all $t \in [n]$ and for all $k \in [K]$,* `Rule` *outputs an arm J_n with the guarantee that its probability of error $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$ verifies*

$$e_{\mathrm{ADV}(\boldsymbol{g})}(n) \le K \exp\left(-\frac{3n}{28 H_{\mathrm{UNIF}}(\boldsymbol{g})}\right).$$

Proof Fix a deterministic oblivious table $\boldsymbol{g} \in [0,1]^{K\times n}$ and relabel its unique best arm as 1. Only the learner is random. Its fresh uniform draws make the selection indicators independent across rounds for each fixed arm k, with Bernoulli parameter $1/K$. Thus $\widetilde{\boldsymbol{g}}_{k,t} = K\boldsymbol{g}_{k,t}\mathbf{1}\{I_t = k\}$. Put

$$X_{k,t} = \widetilde{\boldsymbol{g}}_{k,t} - \boldsymbol{g}_{k,t}, \qquad M_k(n) = \sum_{t=1}^{n} X_{k,t} = \widetilde{G}_{k,n} - G_{k,n},$$

where $G_{k,n} = \sum_{t=1}^{n} \boldsymbol{g}_{k,t}$. For each fixed arm, the centered variables are independent and satisfy

$$\mathbb{E}[X_{k,t}] = 0, \qquad |X_{k,t}| \le K,$$
$$\sigma_{X_{k,t}}^2 = K^2 \boldsymbol{g}_{k,t}^2 \frac{1}{K}\left(1 - \frac{1}{K}\right) = (K-1)\boldsymbol{g}_{k,t}^2 \le K.$$

Hence $\sum_{t=1}^{n} \sigma_{X_{k,t}}^2 \le nK$.

For this table, $n\Delta_j^{\boldsymbol{g}} = G_{1,n} - G_{j,n}$ for $j \neq 1$, and $\Delta_1^{\boldsymbol{g}} = \min_{j\neq 1} \Delta_j^{\boldsymbol{g}}$. The recommendation satisfies the error-event inclusion

$$\{J_n \neq 1\} \subseteq \{M_1(n) \le -n\Delta_1^{\boldsymbol{g}}/2\} \cup \bigcup_{j=2}^{K} \{M_j(n) \ge n\Delta_j^{\boldsymbol{g}}/2\}.$$

Indeed, outside the right-hand side, for every $j \neq 1$,

$$\widetilde{G}_{1,n} > G_{1,n} - n\Delta_1^{\boldsymbol{g}}/2 \ge G_{1,n} - n\Delta_j^{\boldsymbol{g}}/2$$
$$= G_{j,n} + n\Delta_j^{\boldsymbol{g}}/2 > \widetilde{G}_{j,n}.$$

Thus arm 1 is the unique estimated maximizer, so estimated-score ties cause no difficulty.

Bernstein's inequality for the independent centered variables gives, for each sign separately and $z > 0$,

$$\mathbb{P}(\pm M_k(n) \ge z) \le \exp\left(-\frac{z^2}{2(nK + Kz/3)}\right).$$

With $z = n\Delta_k^{\boldsymbol{g}}/2$ and $0 < \Delta_k^{\boldsymbol{g}} \le 1$, this yields

$$\mathbb{P}\left(\pm M_k(n) \ge n\Delta_k^{\boldsymbol{g}}/2\right) \le \exp\left(-\frac{n(\Delta_k^{\boldsymbol{g}})^2}{8K(1 + \Delta_k^{\boldsymbol{g}}/6)}\right)$$
$$\le \exp\left(-\frac{3n(\Delta_k^{\boldsymbol{g}})^2}{28K}\right).$$

A union bound over the K error events, one best-arm lower tail and one upper tail for each competitor, therefore gives

$$e_{\mathrm{ADV}(\boldsymbol{g})}(n) \le \sum_{k=1}^{K} \exp\left(-\frac{3n(\Delta_k^{\boldsymbol{g}})^2}{28K}\right) \le K \exp\left(-\frac{3n(\Delta_1^{\boldsymbol{g}})^2}{28K}\right) = K \exp\left(-\frac{3n}{28H_{\mathrm{UNIF}}(\boldsymbol{g})}\right),$$

using $\Delta_k^{\boldsymbol{g}} \ge \Delta_1^{\boldsymbol{g}}$ and the existing definition $H_{\mathrm{UNIF}}(\boldsymbol{g}) = K/(\Delta_1^{\boldsymbol{g}})^2$. ■

Appendix C. Change of measure

The next lemma is a finite-horizon version of the observed-history change-of-measure method; see Kaufmann et al. (2016, Lemma 1 and Appendix A). We give a self-contained proof allowing common finite-support reward laws and include the Bretagnolle–Huber testing bound; see Lattimore and Szepesvári (2020, Theorem 14.2).

Lemma 10 (Change of measure for observed transcripts) *Fix a positive integer horizon n, $K \ge 2$ arms, an arm $k \in [K]$, and a deterministic set of rounds $\mathcal{T} \subseteq \{1, \dots, n\}$, possibly the entire horizon. The arm and set are fixed before interaction. In each of two environments A and B, the reward-table entries are independent, independent of the learner's internal randomness, and have finite-support distributions in $[0,1]$. At the pairs (k,t) with $t \in \mathcal{T}$, environment A uses Bernoulli(u) and environment B uses Bernoulli(v), with fixed $u, v \in (0,1)$. At every other arm-round pair, the two distributions are identical; these common distributions may depend on the arm and round and may be deterministic. Since there are finitely many entries, a finite reward alphabet contains all their supports in both environments, including any values other than 0 and 1.*

Run the same adaptive randomized learner with the same horizon in both environments. Its actions use only past actions, observed rewards, and internal randomness, which may persist across rounds. It is not given the environment's identity. Its final recommendation $J_n \in [K]$ also uses only the observed interaction and internal randomness. Let $X_t = \boldsymbol{g}_{I_t,t}$ be its observed reward and set

$$Z = (I_1, X_1, \dots, I_n, X_n, J_n).$$

Only observed rewards belong to Z, not the complete table. Let P_A and P_B be its laws on the finite transcript space $\mathcal{Z}$, and let $\mathbb{E}_A$ denote expectation in environment A. Define

$$N_k(\mathcal{T}) = \sum_{t \in \mathcal{T}} \mathbf{1}\{I_t = k\}.$$

Use natural logarithms and binary relative entropy

$$\mathrm{kl}(a,b) = a \log \frac{a}{b} + (1-a) \log \frac{1-a}{1-b}, \qquad a \in [0,1], \quad b \in (0,1),$$

where zero-weight terms are zero. For laws with common support on $\mathcal{Z}$, define

$$D(P_A \| P_B) = \sum_{z : P_A(z) > 0} P_A(z) \log \frac{P_A(z)}{P_B(z)}.$$

Then:

(i) *P_A and P_B have the same support.*

(ii) *Their relative entropy satisfies*

$$D(P_A\|P_B) = \mathbb{E}_A[N_k(\mathcal{T})]\,\mathrm{kl}(u,v).$$

(iii) *For every event W determined by the observed transcript, set $p = P_A(W)$ and $q = P_B(W)$. If $0 < q < 1$, then*

$$\mathrm{kl}(p,q) \le D(P_A\|P_B), \qquad p\log(1/q) \le \mathbb{E}_A[N_k(\mathcal{T})]\,\mathrm{kl}(u,v) + \log 2.$$

If $q = 0$, then $p = 0$; if $q = 1$, then $p = 1$.

(iv) *For every observed-transcript event W, the Bretagnolle–Huber testing bound holds:*

$$P_A(W^c) + P_B(W) \ge \frac{1}{2}\exp\left(-D(P_A\|P_B)\right). \tag{1}$$

The event must be an observed-transcript event; this statement does not directly apply to events involving unobserved table entries.

Proof Draw all learner randomness as a single seed R before interaction, independently of the reward table. Once this seed is fixed, actions, internal state updates, and the final recommendation are determined by the observed history. This includes randomness retained across rounds and final randomization. For a fixed transcript $z = (i_1, x_1, \ldots, i_n, x_n, j)$, let C_z be the set of seeds for which the learner produces the recorded actions and recommendation when sequentially supplied the recorded observations. Set $w(z) = \mathbb{P}_R(C_z)$ and let $f^E_{j,t}(x)$ be the probability mass function of entry (j,t) in environment E on the finite reward alphabet. For each seed in C_z, the observed table coordinates (i_t, t) are distinct and independent. Their probability of matching the recorded rewards is the product of their probability masses; all unobserved entries integrate to one. Integrating over these seeds therefore gives, for $E \in \{A, B\}$,

$$P_E(z) = w(z)\prod_{t=1}^{n} f^E_{i_t,t}(x_t).$$

The factor $w(z)$ is the same in both environments and need not sum to one over transcripts. At each arm-round pair, the two reward distributions have the same support: they are either identical or Bernoulli with parameters in $(0,1)$. Although common reward factors can be zero, the factorization therefore implies

$$P_A(z) > 0 \quad \Longleftrightarrow \quad P_B(z) > 0.$$

This proves common support without requiring the learner to assign positive probability to every action.

On this common positive support, $w(z)$ and all observed reward factors are positive. The learner factor and identical reward factors cancel from the likelihood ratio, leaving only observed pulls of arm k at rounds in $\mathcal{T}$. Thus

$$\log\frac{P_A(Z)}{P_B(Z)} = \sum_{t\in\mathcal{T}} \mathbf{1}\{I_t = k\}\left[X_t\log\frac{u}{v} + (1-X_t)\log\frac{1-u}{1-v}\right].$$

At each round, condition on the seed, the past interaction, and the selected arm. Under A, if $t \in \mathcal{T}$ and $I_t = k$, the current observed reward is Bernoulli(u), since its table entry is independent of the seed and past. Taking expectations yields

$$D(P_A \| P_B) = \mathbb{E}_A \left[\sum_{t \in \mathcal{T}} \mathbf{1}\{I_t = k\} \operatorname{kl}(u, v) \right] = \mathbb{E}_A[N_k(\mathcal{T})] \operatorname{kl}(u, v).$$

Common deterministic rounds contribute zero additional information, even though the learner's actions and final recommendation may depend on their rewards.

For $0 < q < 1$, regard W as a subset of $\mathcal{Z}$ and let $\mathcal{S}$ be the common positive support. Applying the finite log-sum inequality separately to W and its complement gives

$$\begin{aligned} D(P_A \| P_B) &= \sum_{z \in W \cap \mathcal{S}} P_A(z) \log \frac{P_A(z)}{P_B(z)} + \sum_{z \in \mathcal{S} \setminus W} P_A(z) \log \frac{P_A(z)}{P_B(z)} \\ &\geq p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q} = \operatorname{kl}(p, q). \end{aligned}$$

Zero-mass groups are interpreted by continuity. Moreover,

$$\begin{aligned} \operatorname{kl}(p, q) &= p \log(1/q) + (1-p) \log(1/(1-q)) \\ &\quad + p \log p + (1-p) \log(1-p) \\ &\geq p \log(1/q) - \log 2. \end{aligned}$$

The second term in the first line is nonnegative, and binary entropy is maximized at $p = 1/2$ with value $\log 2$. Endpoint values of p use the zero-weight convention. This proves the event consequence. When $q = 0$ or $q = 1$, common support gives respectively $p = 0$ or $p = 1$ without evaluating undefined logarithms.

For the testing bound, on the common positive support $\mathcal{S}$, define

$$a = \sum_{z \in \mathcal{S}} \sqrt{P_A(z) P_B(z)}, \qquad s = \sum_{z \in \mathcal{S}} \min\{P_A(z), P_B(z)\}.$$

Jensen's inequality gives

$$\log a = \log \mathbb{E}_A \left[\sqrt{\frac{P_B(Z)}{P_A(Z)}} \right] \geq -\frac{1}{2} D(P_A \| P_B),$$

so $a^2 \geq \exp(-D(P_A \| P_B))$. By Cauchy–Schwarz,

$$a^2 \leq \left(\sum_{z \in \mathcal{S}} \min\{P_A(z), P_B(z)\} \right) \left(\sum_{z \in \mathcal{S}} \max\{P_A(z), P_B(z)\} \right) = s(2 - s) \leq 2s.$$

Finally, $P_A(W^c) + P_B(W) \geq s$, proving Equation 1. This also covers W empty or equal to the whole transcript space without taking logarithms of event probabilities.

The distinguishing information equals the expected number of informative pulls times the information supplied by each pull. For $\mathcal{T} = \{1, \ldots, m\}$, $N_k(\mathcal{T}) = T_k(m)$. ■

Appendix D. Lower bound in the general adversarial setting

Theorem 2 (Lower bound for the adversarial problem) *Given any problem class $\boldsymbol{\Delta}_3$ with associated complexity H_{UNIF}, for any learner, for any horizon n such that $K\exp\left(-n\Delta_1^2/128\right) \leq 1/128$ and $K \geq 4096$, there exist $\boldsymbol{g}^1 \in \boldsymbol{\Delta}_3$ and $\boldsymbol{g}^2 \in \boldsymbol{\Delta}_3$ so that the probabilities of error suffered by the learner on $\boldsymbol{g}^1$ and $\boldsymbol{g}^2$, denoted $e_{\boldsymbol{g}^1}(n)$ and $e_{\boldsymbol{g}^2}(n)$ respectively, verify*

$$\max\left(e_{\boldsymbol{g}^1}(n), e_{\boldsymbol{g}^2}(n)\right) \geq \min\left(\frac{1}{128}\exp\left(-\frac{32n}{H_{\text{UNIF}}}\right), \frac{1}{32}\right).$$

Proof Fix the reference gaps $0 < \Delta_1 = \Delta_2 \leq \cdots \leq \Delta_K \leq 1/8$, a learner, and a positive integer horizon n satisfying Theorem 2. Write $H = H_{\text{UNIF}} = K/\Delta_1^2$ for the reference complexity. The learner may know the horizon. The constructions and selected arm below may depend on the learner and horizon, but not on its realized random choices. Set

$$m = \lfloor n/2 \rfloor, \qquad \beta = m/n.$$

The horizon assumption gives

$$n\Delta_1^2 \geq 128\log(128K) \geq 1.$$

Since $\Delta_1 \leq 1/8$, we have $n \geq 64$, and therefore $1 \leq m < n$. Moreover,

$$\frac{11}{24} \leq \frac{1}{2} - \frac{1}{2n} \leq \beta \leq \frac{1}{2},$$

where the first inequality uses $n \geq 12$.

Consider a base problem BASE with independent Bernoulli rewards of means $\mu_1^{\text{BASE}} = 1/2$ and $\mu_j^{\text{BASE}} = 1/2 - \Delta_j$ for $j \in [2:K]$. Let $T_j(m)$ be the number of pulls of arm j through round m. Since the expected pull counts of arms in $[2:K]$ sum to at most m, choose a fixed arm $\overline{k} \in [2:K]$ such that

$$\mathbb{E}_{\text{BASE}}\left[T_{\overline{k}}(m)\right] \leq \frac{m}{K-1} \leq \frac{n}{K}. \tag{2}$$

The last inequality follows from $m \leq n/2$ and $K \leq 2(K-1)$ for $K \geq 2$. This arm is chosen from expectations under BASE, before either adversarial table is sampled. The initial phase is the prefix $L = \{1, \ldots, m\}$.

Set

$$c = \Delta_{\overline{k}} - \Delta_1.$$

Since $\Delta_1 \leq \Delta_{\overline{k}} \leq 1/8$, we have $0 \leq c \leq 1/8$. Construct two distributions ADV_1 and ADV_2 over complete reward tables, sampled before interaction and independently of the learner's internal randomness. In ADV_1, the prefix follows BASE on all arms. In ADV_2, the prefix also follows BASE, except that arm $\overline{k}$ has Bernoulli mean $1/2 - \Delta_{\overline{k}} + 2\Delta_1$. All prefix entries are independent across arms and rounds. In both environments the second phase has m rounds and is identical and deterministic: for $m < t \leq 2m$,

$$\boldsymbol{g}_{\overline{k},t} = c, \qquad \boldsymbol{g}_{j,t} = 0 \quad (j \neq \overline{k}).$$

For $2m < t \leq n$, set $\boldsymbol{g}_{j,t} = 0$ for every arm j. Thus even horizons give the original two equal phases, while odd horizons append one all-zero final round. These rewards lie in $[0,1]$. The original

learner, configured for horizon n, chooses an arm in every round, including the optional last round, and recommends an arm only after round n. Let $\mathbb{P}_\ell$ and $\mathbb{E}_\ell$ denote the full table-and-learner probabilities and expectations in ADV_ℓ, for $\ell \in \{1,2\}$. On tables with tied maximizing arms, define $k^\star_{\boldsymbol{g}}$ as the smallest maximizing index.

Write $M_j^\ell = \mathbb{E}_\ell[G_j]$, where $G_j = \sum_{t=1}^n \boldsymbol{g}_{j,t}$. The identity $mc = m(\Delta_{\overline{k}} - \Delta_1)$ gives the exact expected totals below; the optional final zero round contributes nothing:

$$\begin{aligned} M_1^1 &= M_1^2 = m/2, \\ M_j^1 &= M_j^2 = m(1/2 - \Delta_j) \quad (j \notin \{1, \overline{k}\}), \\ M_{\overline{k}}^1 &= m(1/2 - \Delta_{\overline{k}}) + mc = m(1/2 - \Delta_1), \\ M_{\overline{k}}^2 &= m(1/2 - \Delta_{\overline{k}} + 2\Delta_1) + mc = m(1/2 + \Delta_1). \end{aligned}$$

The common deterministic second phase places arm $\overline{k}$ just below arm 1 in the first expected table and just above it in the second. The expected average gap between them is $\beta\Delta_1$, irrespective of the original size of $\Delta_{\overline{k}}$, while the environments differ only in the initial rewards of the lightly sampled arm.

For $\ell \in \{1,2\}$ define the full-table good event

$$\xi_\ell = \left\{ \forall j \in [K], \quad |G_j - M_j^\ell| \le \frac{n\Delta_1}{16} \right\}.$$

Only the first m rewards of each arm are random. Two-sided Hoeffding and a union bound over the complete arm sequences give

$$\mathbb{P}_\ell(\xi_\ell^c) \le 2K \exp\left(-\frac{2(n\Delta_1/16)^2}{m}\right) \le 2K \exp\left(-\frac{n\Delta_1^2}{64}\right), \tag{3}$$

since $m \le n/2$. These events involve the full reward table and are generally not determined by the learner's observations.

On either good event, every normalized pairwise difference deviates from its expectation by at most $\Delta_1/8$. On ξ_1, the expected differences from arm 1 are $\beta\Delta_1$ for arm $\overline{k}$ and $\beta\Delta_j$ for $j \notin \{1, \overline{k}\}$. Since $\beta - 1/8 \ge 1/3$, $\beta + 1/8 \le 5/8$, and $\Delta_1 \le \Delta_j$,

$$\frac{\Delta_1}{3} \le \frac{G_1 - G_{\overline{k}}}{n} \le \frac{5\Delta_1}{8}, \qquad \frac{\Delta_j}{3} \le \frac{G_1 - G_j}{n} \le \frac{5\Delta_j}{8} \quad (j \notin \{1, \overline{k}\}).$$

All these differences are positive, so arm 1 is uniquely best. Its own gap is the minimum of these differences; the lower bounds and the upper bound from arm $\overline{k}$ give

$$\frac{\Delta_1}{3} \le \Delta_1^{\boldsymbol{g}} \le \frac{5\Delta_1}{8}.$$

Thus every label satisfies the defining bounds for $\boldsymbol{\Delta}_3$, with $\overline{k}$ as the exceptional label, and every table in ξ_1 belongs to $\boldsymbol{\Delta}_3$.

On ξ_2, the expected differences from arm $\overline{k}$ are $\beta\Delta_1$ for arm 1 and $\beta(\Delta_j + \Delta_1)$ for $j \notin \{1, \overline{k}\}$. Consequently,

$$\frac{\Delta_1}{3} \le \frac{G_{\overline{k}} - G_1}{n} \le \frac{5\Delta_1}{8}, \qquad \frac{\Delta_j}{3} \le \frac{G_{\overline{k}} - G_j}{n} \le \frac{9\Delta_j}{8} \quad (j \notin \{1, \overline{k}\}).$$

For the latter lower bound, $\beta(\Delta_j + \Delta_1) - \Delta_1/8 \geq (\beta - 1/8)\Delta_j \geq \Delta_j/3$. For the upper bound, $\beta \leq 1/2$ and $\Delta_1 \leq \Delta_j$ give $\beta(\Delta_j + \Delta_1) + \Delta_1/8 \leq 9\Delta_j/8$. Arm $\overline{k}$ is uniquely best, and its own gap, the minimum of its differences from every other arm, satisfies

$$\frac{\Delta_1}{3} \leq \Delta_{\overline{k}}^{\boldsymbol{g}} \leq \frac{5\Delta_1}{8}.$$

The upper bound uses the difference from arm 1. Label 1 meets its ordinary reference-gap requirement, all other labels except $\overline{k}$ meet theirs, and $\overline{k}$ meets the exceptional requirement. Every table in ξ_2 therefore also belongs to $\boldsymbol{\Delta}_3$.

Let P_1, P_2 be the observed-transcript laws from Lemma 10 under $\mathrm{ADV}_1, \mathrm{ADV}_2$, distinguished from the full laws $\mathbb{P}_1, \mathbb{P}_2$. Apply the lemma with $A = \mathrm{ADV}_1$, $B = \mathrm{ADV}_2$, $\mathcal{T} = \{1, \ldots, m\}$, and

$$u = \frac{1}{2} - \Delta_{\overline{k}}, \qquad v = \frac{1}{2} - \Delta_{\overline{k}} + 2\Delta_1.$$

All table entries are independent of one another and of the learner's randomness. The common distributions have finite support, including the identical deterministic second phase and optional last round, and $u, v \in [3/8, 5/8] \subset (0, 1)$. This entire continuation contributes no additional likelihood-ratio factors. The same learner is run with the same horizon. The lemma gives

$$D(P_1 \| P_2) = \mathbb{E}_1[T_{\overline{k}}(m)]\,\mathrm{kl}(u, v) = \mathbb{E}_{\mathrm{BASE}}[T_{\overline{k}}(m)]\,\mathrm{kl}(u, v).$$

The second equality holds because ADV_1 and BASE have identical interaction laws through the prefix, including the learner's adaptive actions. No law is conditioned on a good-table event.

Using $\log z \leq z - 1$,

$$\mathrm{kl}(u, v) \leq u\left(\frac{u}{v} - 1\right) + (1 - u)\left(\frac{1 - u}{1 - v} - 1\right) = \frac{(u - v)^2}{v(1 - v)}.$$

Here $|u - v| = 2\Delta_1$ and $v(1 - v) \geq 15/64$, so $\mathrm{kl}(u, v) \leq (256/15)\Delta_1^2$. Set $x = n/H = n\Delta_1^2/K > 0$. Equation 2 yields

$$D(P_1 \| P_2) \leq \frac{256}{15}\Delta_1^2 \frac{n}{K} = \frac{256}{15}x \leq 32x. \tag{4}$$

For the transcript event $W = \{J_n = 1\}$, the testing bound in Equation 1 now gives

$$P_1(J_n \neq 1) + P_2(J_n = 1) \geq \frac{1}{2}\exp(-D(P_1 \| P_2)) \geq \frac{1}{2}\exp(-32x).$$

The learner may recommend any arm; recommending arm 1 is simply one error event in the second environment on its good tables.

Write $\varepsilon = \exp(-32x)$ and $b_\ell = \mathbb{P}_\ell(\xi_\ell^c)$. Both theorem assumptions are needed to compare the bad-table probabilities with this testing bound. Equation 3 and $K\exp(-n\Delta_1^2/128) \leq 1/128$ give

$$\begin{aligned} b_\ell &\leq 2K\exp(-n\Delta_1^2/64) \\ &= 2\left[K\exp(-n\Delta_1^2/128)\right]\exp(-n\Delta_1^2/128) \\ &\leq \frac{1}{64}\exp(-n\Delta_1^2/128) \leq \frac{\varepsilon}{64}. \end{aligned}$$

The last inequality uses $K \geq 4096$, which implies $32/K \leq 1/128$. Thus bad tables cost only a fixed fraction of the exponentially small testing bound.

For a fixed table $\boldsymbol{g}$, let $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$ denote error over the learner's internal randomness, using the best-arm convention above. Define its error mass on good tables by

$$R_\ell = \mathbb{E}_{\boldsymbol{g}\sim\mathrm{ADV}_\ell}\left[e_{\mathrm{ADV}(\boldsymbol{g})}(n)\mathbf{1}\{\boldsymbol{g}\in\xi_\ell\}\right].$$

The unique best arms on the good events give

$$\begin{aligned} R_1 &= \mathbb{P}_1(\{J_n \neq 1\}\cap\xi_1), \\ R_2 &= \mathbb{P}_2(\{J_n \neq \overline{k}\}\cap\xi_2) \geq \mathbb{P}_2(\{J_n = 1\}\cap\xi_2). \end{aligned}$$

Subtracting the bad-table probabilities therefore yields

$$\begin{aligned} R_1 + R_2 &\geq P_1(J_n \neq 1) + P_2(J_n = 1) - b_1 - b_2 \\ &\geq \frac{\varepsilon}{2} - \frac{2\varepsilon}{64} = \frac{15\varepsilon}{32}. \end{aligned} \tag{5}$$

All joint events involving ξ_ℓ use the full law $\mathbb{P}_\ell$; no independence from the recommendation is required.

Each table distribution has finite support: only its mK prefix entries are random Bernoulli variables and its suffix is fixed. Each good subset of that support is nonempty, since $b_\ell \leq \varepsilon/64 < 1$. Choose a deterministic table $\boldsymbol{g}^\ell$ maximizing $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$ over the good support of ADV_ℓ. Then

$$R_\ell \leq \mathbb{P}_\ell(\xi_\ell)e_{\mathrm{ADV}(\boldsymbol{g}^\ell)}(n) \leq e_{\mathrm{ADV}(\boldsymbol{g}^\ell)}(n).$$

Equation 5 implies

$$\max_{\ell\in\{1,2\}} e_{\mathrm{ADV}(\boldsymbol{g}^\ell)}(n) \geq \frac{R_1+R_2}{2} \geq \frac{15}{64}\exp\left(-\frac{32n}{H}\right).$$

Both tables belong to $\boldsymbol{\Delta}_3$ and have different unique best arms, 1 and $\overline{k}$. Since $15/64 \geq 1/128$, this bound implies

$$\max_{\ell\in\{1,2\}} e_{\boldsymbol{g}^\ell}(n) \geq \min\left\{\frac{1}{128}\exp\left(-\frac{32n}{H_{\mathrm{UNIF}}}\right), \frac{1}{32}\right\},$$

which is the claimed conclusion. The extraction uses finite-support maximization and requires neither conditional probabilities on null events nor a coupling of the two table draws. ■

Appendix E. Lower bound in the best of both worlds

Theorem 4 (Lower bound for the BOB challenge) *Fix $K \geq 2$ and reference gaps $0 < \Delta_1 = \Delta_2 \leq \cdots \leq \Delta_K \leq 1/8$, and let $\boldsymbol{\Delta}_4$ be the corresponding problem class. For any learner and any positive integer horizon n satisfying*

$$K\exp\left(-\frac{\Delta_1^2 n}{128}\right) \leq \frac{1}{32},$$

there exists an i.i.d. stochastic problem $\mathrm{STO} \in \boldsymbol{\Delta}_4$ *such that, if*

$$e_{\mathrm{STO}}(n) \le \frac{1}{64}\exp\left(-\frac{2048n}{H^{\mathrm{STO}}_{\mathrm{BOB}}}\right),$$

then there exists a deterministic gain table $\boldsymbol{g} \in [0,1]^{K\times n}$ *belonging to* $\boldsymbol{\Delta}_4$ *for which*

$$e_{\mathrm{ADV}(\boldsymbol{g})}(n) \ge \frac{1}{16}.$$

Here $H^{\mathrm{STO}}_{\mathrm{BOB}}$ *is the complexity computed from the gaps of the constructed stochastic problem. The constructed stochastic problem and gain table may depend on the learner and on* n*.*

Proof Fix the reference gaps defining $\boldsymbol{\Delta}_4$, a learner, and a horizon n satisfying the hypothesis of the theorem. Throughout the proof, the learner is run with this fixed horizon, which it may know. The constructions below may depend on this learner and on n. Let $T_k(t)$ be the number of times arm k has been pulled by the end of round t. Let us consider a base problem, denoted BASE, with $\nu_1,\dots,\nu_K$, Bernoulli distributions with mean $\mu_1^{\mathrm{BASE}},\dots,\mu_K^{\mathrm{BASE}}$ such that $\mu_1^{\mathrm{BASE}} \triangleq 1/2$, and for all $k \in [2,K]$, $\mu_k^{\mathrm{BASE}} \triangleq 1/2 - \Delta_k$ (the gaps specified in the claim of the theorem). The expectation and probability with respect to the learner and the samples of this problem are $\mathbb{E}_{\mathrm{BASE}}$ and $\mathbb{P}_{\mathrm{BASE}}$.

With respect to this base problem, choose any maximizing index $i \in \arg\max_{j\in[K]} \frac{j}{\Delta_j}$. **Choice of the initial phase length.** We choose the length of the initial phase so as to maximize the resulting lower bound on the learner's error. The stochastic and adversarial constructions differ only during this phase. A shorter phase reduces the phase-length contribution $n_i\Delta_i^2/i$ to the information-theoretic estimate and therefore strengthens the resulting error lower bound.

However, the initial phase must be sufficiently long to ensure that the adversarial problem belongs to the prescribed class $\boldsymbol{\Delta}_4$. Its expected gaps must remain large enough after accounting for fluctuations of the cumulative rewards.

We therefore seek the shortest initial phase, up to universal constant factors, that guarantees these gap constraints uniformly over arms satisfying the selection bounds. The choice below uses a convenient factor 2 to provide additional gap margin. Together with the phase bound below, it keeps $n_i\Delta_i^2/i$ bounded by a universal constant times $n/H^{\mathrm{BASE}}_{\mathrm{BOB}}$.

Set

$$a_i \triangleq \min\left\{1, \frac{2\Delta_1}{\Delta_i}\right\}, \qquad 0 < a_i \le 1.$$

The initial phase L_i is the prefix of rounds $t = 1,\dots,n_i$, where

$$n_i \triangleq \lceil na_i \rceil = \min\left\{n, \left\lceil \frac{2n\Delta_1}{\Delta_i} \right\rceil\right\}.$$

The equality uses the fact that n is an integer; in particular, $1 \le n_i \le n$.

The theorem's horizon condition implies

$$n\Delta_1^2 \ge 128\log(32K) \ge 1.$$

Writing $r = n\Delta_1/\Delta_i$, since $\Delta_1\Delta_i \le 1/64$, we have

$$r = \frac{n\Delta_1^2}{\Delta_1\Delta_i} \ge 1, \qquad \frac{\Delta_i}{n\Delta_1} = \frac{\Delta_1\Delta_i}{n\Delta_1^2} \le \frac{1}{64} \le \frac{1}{8}.$$

Thus $\Delta_i/n \le \Delta_1/8$, and the capped ceiling gives

$$n_i \le \lceil 2r \rceil \le 2r + 1 \le 3r = \frac{3n\Delta_1}{\Delta_i}.$$

Since $\Delta_1 = \Delta_2 > 0$, we have $1/\Delta_1 < 2/\Delta_2$. Thus $i \ge 2$ for every choice of maximizing index.

Now we want to identify an arm with a gap of the same order as Δ_i that the learner will not pull very much in the base problem BASE during the period L_i. This arm will be the one 'attacked' by the adversary to confuse the learner as the learner is not pulling it very much.

If $i = 2$, choose $\overline{k} = 2$. Since at most one pull of arm 2 is made in each round, $T_2(n_i) \le n_i$ almost surely, and hence

$$\mathbb{E}_{\text{BASE}}\left[T_2(n_i)\right] \le n_i = \frac{2n_i}{i}.$$

Under the phase definition above, $i = 2$ gives $a_i = 1$ and $n_i = n$. The second phase is then empty, and the ADV construction below coincides with BASE throughout the horizon. This empty second phase is allowed by the construction.

If $i \ge 3$, consider the arms with indices in $[\lceil i/2 \rceil : i]$. There are $\lfloor i/2 \rfloor + 1 > i/2$ such arms. Since

$$\sum_{k \in [K]} \mathbb{E}_{\text{BASE}}\left[T_k(n_i)\right] = n_i,$$

they cannot all have expected pull counts strictly greater than $2n_i/i$. Choose any $\overline{k} \in [\lceil i/2 \rceil : i]$ whose expected pull count is at most $2n_i/i$. In this case $\overline{k} \ge \lceil i/2 \rceil \ge 2$.

By construction we have $\Delta_{\overline{k}} \le \Delta_i$ and

$$\mathbb{E}_{\text{BASE}}\left[T_{\overline{k}}(n_i)\right] \le \frac{2n_i}{i}. \tag{6}$$

In both cases, $2 \le \overline{k} \le i$ and $\overline{k} \ge i/2$. The maximizing property of i gives $i/\Delta_i \ge \overline{k}/\Delta_{\overline{k}}$, so

$$\Delta_i \le \frac{i}{\overline{k}}\Delta_{\overline{k}} \le 2\Delta_{\overline{k}}. \tag{7}$$

Equality is permitted in this gap comparison. The resulting bound $\Delta_{\overline{k}}/\Delta_i \ge 1/2$ is sufficient for the expected-gap lower bound below.

Now we construct an i.i.d. stochastic problem, denoted STO, where the arm distributions are the same as in the BASE problem, except for arm $\overline{k}$, $\mu_k^{\text{STO}} \triangleq \mu_k^{\text{BASE}}$ for all $k \ne \overline{k}$. We set in STO, $\mu_{\overline{k}}^{\text{STO}} \triangleq 1/2 + \Delta_1/4$.

The best arm in the constructed STO problem is $k^\star_{\text{STO}} \triangleq \overline{k}$. Its gaps, indexed by the original arm labels, satisfy

$$\Delta_1^{\text{STO}} = \Delta_{\overline{k}}^{\text{STO}} = \frac{\Delta_1}{4},$$

and, for every $j \in [K] \setminus \{1, \overline{k}\}$,

$$\Delta_j \le \Delta_j^{\text{STO}} = \Delta_j + \frac{\Delta_1}{4} \le \frac{5}{4}\Delta_j.$$

Thus label 1 has gap $\Delta_1/4$, the selected label $\overline{k}$ is the exceptional label with the same gap, and every remaining label satisfies the ordinary bounds $[\Delta_j/4, 4\Delta_j]$. Consequently, STO $\in \boldsymbol{\Delta}_4$.

We next compare the complexity of the constructed stochastic problem with that of BASE. Since the construction changes the ordering of the arms, this comparison must use the sorted stochastic gaps. Let $h_1 \le \cdots \le h_K$ be the sorted values of the labelled gaps Δ_j^{STO}, and define the local quantity

$$Q \triangleq \max_{j\in[K]} \frac{j}{\Delta_j}, \qquad H_{\mathrm{BOB}}^{\mathrm{BASE}} = \frac{Q}{\Delta_1}, \qquad Q \ge \frac{2}{\Delta_1}.$$

The sorted stochastic gaps consist of two copies of $\Delta_1/4$, followed by the ordered list $\Delta_j + \Delta_1/4$ for $j \in [2:K] \setminus \{\overline{k}\}$. Deleting the entry for $\overline{k}$ from the ordered list with labels $2, \dots, K$ moves a remaining position by at most one. This comparison remains valid when gaps tie. Hence

$$h_1 = h_2 = \frac{\Delta_1}{4}, \qquad \Delta_{r-1} + \frac{\Delta_1}{4} \le h_r \le \Delta_r + \frac{\Delta_1}{4} \quad (3 \le r \le K).$$

When $K = 2$, there are no ranks $r \ge 3$; only the first two ranks are used below.

For the upper bound, the first two ranks satisfy

$$\max_{r\in\{1,2\}} \frac{r}{h_r} = \frac{8}{\Delta_1} \le 4Q.$$

For each rank $r \ge 3$, we have

$$\frac{r}{h_r} \le \frac{r}{\Delta_{r-1}} = \frac{r}{r-1}\frac{r-1}{\Delta_{r-1}} \le \frac{3}{2}Q \le 4Q.$$

Thus, bounding every rank in $[K]$ gives

$$H_{\mathrm{BOB}}^{\mathrm{STO}} = \frac{4}{\Delta_1} \max_{r\in[K]} \frac{r}{h_r} \le \frac{4}{\Delta_1}(4Q) = 16 H_{\mathrm{BOB}}^{\mathrm{BASE}}.$$

For the lower bound, $h_r \le 2\Delta_r$ for every rank r: for $r = 1, 2$ this follows from $h_r = \Delta_1/4$ and $\Delta_1 = \Delta_2$, while for $r \ge 3$ it follows from $h_r \le \Delta_r + \Delta_1/4$ and $\Delta_1 \le \Delta_r$. Consequently,

$$H_{\mathrm{BOB}}^{\mathrm{STO}} \ge \frac{4}{\Delta_1} \max_{r\in[K]} \frac{r}{2\Delta_r} = 2H_{\mathrm{BOB}}^{\mathrm{BASE}} \ge \frac{1}{2} H_{\mathrm{BOB}}^{\mathrm{BASE}}.$$

Combining these bounds, we retain the weaker lower bound in

$$H_{\mathrm{BOB}}^{\mathrm{BASE}}/2 \le H_{\mathrm{BOB}}^{\mathrm{STO}} \le 16 H_{\mathrm{BOB}}^{\mathrm{BASE}} \tag{8}$$

Writing $H_{\mathtt{SR}}^{\mathrm{BASE}}$ and $H_{\mathtt{SR}}^{\mathrm{STO}}$ for the respective SR complexities, the same sorted-gap comparison and the stronger lower bound above give

$$H_{\mathtt{SR}}^{\mathrm{STO}} \le \max\left\{\frac{32}{\Delta_1^2}, \frac{3}{2} H_{\mathtt{SR}}^{\mathrm{BASE}}\right\} \le 16 H_{\mathtt{SR}}^{\mathrm{BASE}},$$
$$\frac{H_{\mathrm{BOB}}^{\mathrm{STO}}}{H_{\mathtt{SR}}^{\mathrm{STO}}} \ge \frac{1}{8}\frac{H_{\mathrm{BOB}}^{\mathrm{BASE}}}{H_{\mathtt{SR}}^{\mathrm{BASE}}}.$$

For the square-root profile $\Delta_r = \Delta_1\sqrt{r/2}$, $r \in [2:K]$, the ratio for STO is at least $\frac{1}{8}\sqrt{K/2}$, for every eligible choice of arm $\overline{k}$.

The expectation and probability with respect to the learner and the samples of this problem are denoted $\mathbb{E}_{\text{STO}}$ and $\mathbb{P}_{\text{STO}}$.

The second bandit problem is the adversarial one, denoted ADV. Before interaction with the learner, it samples a table $\boldsymbol{g} \in \{0,1\}^{K\times n}$. For any sampled table with several maximizing arms, let $k^\star_{\boldsymbol{g}}$ be the smallest maximizing index. On ξ the maximizer is unique, so this convention does not affect the extracted instance. The Bernoulli entries are independent across arms and rounds, and independent of the learner's internal randomness. The table-generating distribution is fixed once the learner, horizon, reference gaps, and the choices of i and $\overline{k}$ are fixed; it does not adapt to the learner's realized actions. For each $t \in [n]$ and arm k, the (k,t) entry is Bernoulli with mean $\mu_k^{\text{ADV}}(t)$. For all t and $k \neq \overline{k}$, ADV follows the BASE problem: $\mu_k^{\text{ADV}}(t) \triangleq \mu_k^{\text{BASE}}$. For arm $\overline{k}$, until the end of phase L_i, for all t with $1 \le t \le n_i$, $\mu_{\overline{k}}^{\text{ADV}}(t) \triangleq \mu_{\overline{k}}^{\text{BASE}}$ and then a switch happens, for $n_i < t \le n$, arm $\overline{k}$ possesses the same distributions as in the STO problem, $\mu_{\overline{k}}^{\text{ADV}}(t) \triangleq \mu_{\overline{k}}^{\text{STO}}$. The expectation and probability with respect to the learner and the samples of this problem are denoted $\mathbb{E}_{\text{ADV}}$ and $\mathbb{P}_{\text{ADV}}$.

Let us compute the expected cumulative gains for this fixed construction. For each arm $k \in [K]$, let $M_k \triangleq \sum_{t=1}^n \mu_k^{\text{ADV}}(t)$. Then

$$M_{\overline{k}} = n_i\left(\frac{1}{2} - \Delta_{\overline{k}}\right) + (n-n_i)\left(\frac{1}{2} + \frac{\Delta_1}{4}\right), \qquad M_1 = \frac{n}{2},$$

and, for $j \in [K] \setminus \{1, \overline{k}\}$,

$$M_j = n\left(\frac{1}{2} - \Delta_j\right).$$

Define the expected average gap

$$b \triangleq \frac{M_1 - M_{\overline{k}}}{n} = \frac{n_i}{n}\Delta_{\overline{k}} - \left(1 - \frac{n_i}{n}\right)\frac{\Delta_1}{4}.$$

If $n_i = n$, then $b = \Delta_{\overline{k}} \ge \Delta_1$. This case also includes a full-horizon phase caused by ceiling rounding. If $n_i < n$, its definition gives $n_i \ge 2n\Delta_1/\Delta_i$, so Equation 7 yields

$$b \ge \frac{2\Delta_1\Delta_{\overline{k}}}{\Delta_i} - \frac{\Delta_1}{4} \ge \frac{3\Delta_1}{4}.$$

For the upper bound, in all cases,

$$b \le \frac{n_i}{n}\Delta_{\overline{k}} \le \left(\frac{2\Delta_1}{\Delta_i} + \frac{1}{n}\right)\Delta_i = 2\Delta_1 + \frac{\Delta_i}{n} \le \frac{17\Delta_1}{8}.$$

Consequently,

$$\frac{3\Delta_1}{4} \le b \le \frac{17\Delta_1}{8}.$$

Let us consider the event $\xi = \{\forall k \in [K], |G_k - M_k| \le n\Delta_1/16\}$. For a fixed arm j, G_j is the sum of all n entries in its row of the reward table. These entries are independent and lie in $[0,1]$,

although their means may differ across rounds. This is the full table sum, including rounds on which the learner does not pull arm j. Two-sided Hoeffding gives

$$\begin{aligned}\mathbb{P}_{\mathrm{ADV}}\left(|G_j - M_j| > \frac{n\Delta_1}{16}\right) &\le 2\exp\left(-\frac{2(n\Delta_1/16)^2}{n}\right)\\ &= 2\exp\left(-\frac{n\Delta_1^2}{128}\right).\end{aligned}$$

Therefore, a union bound over arms yields

$$\mathbb{P}_{\mathrm{ADV}}(\xi^c) \le 2K\exp\left(-\frac{n\Delta_1^2}{128}\right) \le \frac{1}{16}. \tag{9}$$

The last inequality uses the theorem hypothesis

$$K\exp\left(-\frac{n\Delta_1^2}{128}\right) \le \frac{1}{32}.$$

Consequently, $\mathbb{P}_{\mathrm{ADV}}(\xi) \ge 15/16$.

On ξ, each difference $G_1 - G_j$ deviates from $M_1 - M_j$ by at most $n\Delta_1/8$. For the selected arm, the bounds on b therefore give

$$\frac{5\Delta_1}{8} \le \frac{G_1 - G_{\overline{k}}}{n} \le \frac{9\Delta_1}{4}.$$

For every $j \in [K] \setminus \{1, \overline{k}\}$, we have

$$\Delta_j - \frac{\Delta_1}{8} \le \frac{G_1 - G_j}{n} \le \Delta_j + \frac{\Delta_1}{8}.$$

Since $\Delta_j \ge \Delta_1$, this implies

$$\frac{7}{8}\Delta_j \le \frac{G_1 - G_j}{n} \le \frac{9}{8}\Delta_j.$$

All these differences are strictly positive, so arm 1 is the unique best arm on ξ. Thus the displayed differences are the gaps $\Delta_j^{\boldsymbol{g}}$ for $j \ne 1$. The best arm's own gap is

$$\Delta_1^{\boldsymbol{g}} = \min_{j \in [K]\setminus\{1\}} \frac{G_1 - G_j}{n}.$$

The lower bounds for all other arms and the upper bound for the selected arm give

$$\frac{5\Delta_1}{8} \le \Delta_1^{\boldsymbol{g}} \le \frac{9\Delta_1}{4}.$$

The bounds for labels 1 and $\overline{k}$ lie in $[\Delta_1/4, 4\Delta_1]$, and the bounds for every remaining label j lie in $[\Delta_j/4, 4\Delta_j]$. Hence every table $\boldsymbol{g} \in \xi$ belongs to $\boldsymbol{\Delta}_4$, with $\overline{k}$ as the exceptional label in the class definition.

Note that therefore the only difference between the two problems STO and ADV is for arm $\overline{k}$ during phase L_i.

The stochastic problem has best arm $\overline{k}$, whereas arm 1 is uniquely best in the adversarial construction on ξ. We compare the distributions of the learner's observations to quantify the difficulty of distinguishing these problems.

Let P_A and P_S denote the laws of the observed transcript Z from Lemma 10 under randomized ADV and STO, respectively. These transcript laws are distinguished from the full table-and-learner law $\mathbb{P}_{\text{ADV}}$ used for ξ, which also involves unobserved table entries. Both constructions have independent Bernoulli entries, independent of the learner's internal randomness, with all parameters in $[3/8, 17/32] \subset (0, 1)$. The same learner is run with the same horizon, and the only differing arm-round pairs are arm $\overline{k}$ through the initial prefix of n_i rounds. The arm and prefix are fixed before interaction. Apply the lemma with $A = \text{ADV}$, $B = \text{STO}$, $\mathcal{T} = \{1, \dots, n_i\}$, and

$$u = \frac{1}{2} - \Delta_{\overline{k}}, \qquad v = \frac{1}{2} + \frac{\Delta_1}{4}.$$

It gives

$$D(P_A \| P_S) = \mathbb{E}_{\text{ADV}}\left[T_{\overline{k}}(n_i)\right] \operatorname{kl}(u, v).$$

Through this fixed prefix, ADV and BASE have identical reward laws and use the same learner. Their interaction laws agree through n_i, including the learner's adaptive actions. Consequently,

$$\mathbb{E}_{\text{ADV}}\left[T_{\overline{k}}(n_i)\right] = \mathbb{E}_{\text{BASE}}\left[T_{\overline{k}}(n_i)\right],$$

and hence

$$D(P_A \| P_S) = \mathbb{E}_{\text{BASE}}\left[T_{\overline{k}}(n_i)\right] \operatorname{kl}(u, v).$$

For $u, v \in (0, 1)$, the inequality $\log z \le z - 1$ gives

$$\begin{aligned} \operatorname{kl}(u, v) &\le u\left(\frac{u}{v} - 1\right) + (1 - u)\left(\frac{1 - u}{1 - v} - 1\right) \\ &= \frac{(u - v)^2}{v(1 - v)}. \end{aligned}$$

Here $\Delta_1 \le \Delta_{\overline{k}} \le 1/8$, so

$$|u - v| \le \frac{5}{4}\Delta_{\overline{k}}, \qquad v(1 - v) = \frac{1}{4} - \frac{\Delta_1^2}{16} \ge \frac{255}{1024}.$$

Thus

$$\operatorname{kl}(u, v) \le \frac{1600}{255}\Delta_{\overline{k}}^2 \le 8\Delta_{\overline{k}}^2.$$

Using Equation 6, $\Delta_{\overline{k}} \le \Delta_i$, the phase-length bound, and $H_{\text{BOB}}^{\text{BASE}} = i/(\Delta_1 \Delta_i)$, we obtain

$$\begin{aligned} D(P_A \| P_S) &\le 8\Delta_{\overline{k}}^2 \frac{2n_i}{i} \\ &\le \frac{16\Delta_i^2 n_i}{i} \\ &\le \frac{48n\Delta_1\Delta_i}{i} \\ &= \frac{48n}{H_{\text{BOB}}^{\text{BASE}}} \\ &\le \frac{768n}{H_{\text{BOB}}^{\text{STO}}}. \end{aligned} \tag{10}$$

The last inequality uses Equation 8: $H^{\mathrm{STO}}_{\mathrm{BOB}} \le 16H^{\mathrm{BASE}}_{\mathrm{BOB}}$ implies $1/H^{\mathrm{BASE}}_{\mathrm{BOB}} \le 16/H^{\mathrm{STO}}_{\mathrm{BOB}}$. The chosen phase is long enough to satisfy the adversarial gap constraints, while its length bound keeps this information cost at most a universal constant times $n/H^{\mathrm{STO}}_{\mathrm{BOB}}$.

Assume the stochastic-error premise of Theorem 4, and set

$$x = \frac{n}{H^{\mathrm{STO}}_{\mathrm{BOB}}} > 0, \qquad p = P_A(J_n = 1), \qquad q = P_S(J_n = 1).$$

Since the unique best arm in STO is $\overline{k} \neq 1$,

$$q \le e_{\mathrm{STO}}(n) \le \frac{1}{64}\exp(-2048x).$$

If $q = 0$, common support in Lemma 10 gives $p = 0$. Otherwise $0 < q < 1$, and its event consequence for $W = \{J_n = 1\}$, together with Equation 10, gives

$$p\log(1/q) \le D(P_A\|P_S) + \log 2 \le 768x + \log 2.$$

The stochastic guarantee also gives $\log(1/q) \ge 2048x + \log 64 = 2048x + 6\log 2$. Therefore,

$$p \le \frac{768x + \log 2}{2048x + 6\log 2} \le \frac{3}{8}.$$

The denominator is positive, $768 = (3/8)2048$, and $\log 2 \le (3/8)6\log 2$, which proves the last inequality. The same conclusion holds when $q = 0$. This bounds the probability of recommending arm 1; that arm need not be best on tables outside ξ.

Return now to the full table-and-learner probability law, and write

$$e_{\mathrm{ADV}}(n) = \mathbb{P}_{\mathrm{ADV}}\left(J_n \neq k^\star_{\boldsymbol{g}}\right),$$

with the fixed table tie-breaking convention above. On ξ, arm 1 is uniquely best, and $\mathbb{P}_{\mathrm{ADV}}(J_n = 1) = p$ because this is a transcript event. Equation 9 therefore yields

$$\begin{aligned} e_{\mathrm{ADV}}(n) &\ge \mathbb{P}_{\mathrm{ADV}}\left(\{J_n \neq 1\} \cap \xi\right) \\ &\ge \mathbb{P}_{\mathrm{ADV}}(J_n \neq 1) - \mathbb{P}_{\mathrm{ADV}}(\xi^c) \\ &= 1 - p - \mathbb{P}_{\mathrm{ADV}}(\xi^c) \\ &\ge 1 - \frac{3}{8} - \frac{1}{16} \\ &= \frac{9}{16} \ge \frac{1}{8}. \end{aligned} \tag{11}$$

No independence between the recommendation and ξ is needed.

Equation 11 gives the lower bound needed for the extraction of a deterministic table:

$$e_{\mathrm{ADV}}(n) \ge 1/8.$$

For a fixed table $\boldsymbol{g}$, write

$$e_{\boldsymbol{g}}(n) \triangleq e_{\mathrm{ADV}(\boldsymbol{g})}(n),$$

where the probability is over the learner's internal randomness. The random ADV model is a distribution over fixed tables, independent of that randomness, so

$$e_{\mathrm{ADV}}(n) = \mathbb{E}_{\boldsymbol{g}\sim\mathrm{ADV}}\left[e_{\boldsymbol{g}}(n)\right].$$

Suppose for contradiction that $e_{\boldsymbol{g}}(n) < 1/16$ for every $\boldsymbol{g} \in \xi$. Splitting this expectation using indicators and applying Equation 9 gives

$$\begin{aligned} e_{\mathrm{ADV}}(n) &= \mathbb{E}_{\boldsymbol{g}\sim\mathrm{ADV}}\left[e_{\boldsymbol{g}}(n)\mathbf{1}\{\boldsymbol{g}\in\xi\}\right] + \mathbb{E}_{\boldsymbol{g}\sim\mathrm{ADV}}\left[e_{\boldsymbol{g}}(n)\mathbf{1}\{\boldsymbol{g}\notin\xi\}\right] \\ &\le \frac{1}{16}\mathbb{P}_{\mathrm{ADV}}(\xi) + \mathbb{P}_{\mathrm{ADV}}(\xi^c) \\ &= \frac{1}{16} + \frac{15}{16}\mathbb{P}_{\mathrm{ADV}}(\xi^c) \\ &\le \frac{1}{16} + \frac{15}{256} \\ &= \frac{31}{256} < \frac{1}{8}. \end{aligned}$$

This contradicts the stated input $e_{\mathrm{ADV}}(n) \ge 1/8$. Hence there exists a deterministic table $\boldsymbol{g} \in \xi$ with $e_{\mathrm{ADV}(\boldsymbol{g})}(n) \ge 1/16$. The gap calculation above shows that this table belongs to $\boldsymbol{\Delta}_4$. ■

Appendix F. Upper bound in best of both worlds

Stochastic setup. We work in the bounded i.i.d. stochastic setting of Section 4, with arm laws ν_h on $[0,1]$ and means μ_h. For the analysis, relabel the arms so that $\mu_1 > \mu_2 \ge \cdots \ge \mu_K$, ordering equal suboptimal means arbitrarily and consistently. Thus $0 < \Delta_1 = \Delta_2 \le \cdots \le \Delta_K \le 1$. This does not affect the learner's rankings or tie resolution. Fix any deterministic admissible schedule $\boldsymbol{a} \in \boldsymbol{A}$ and write $n_r = na_r$, so that

$$n_1 = n_2 = n \ge n_3 \ge \cdots \ge n_K \ge 1, \qquad n_{K+1} = 0.$$

All comparison times are integers. The schedule may depend on the instance and horizon, but not on the realized observations. Coincident comparison times are allowed, and sums over empty time intervals are zero.

For $u \in [n]$, the rank $\widetilde{\langle k\rangle}_u$ is the learner's ranking from $\widetilde{G}_{\cdot,u-1}$. At the unplayed time $n+1$, use a decreasing-score ranking of $\widetilde{G}_{\cdot,n}$ that places the actual recommendation J_n first among tied maximizers; order other ties arbitrarily. This is only an analysis convention: no additional pull is made, and neither the recommendation nor its tie rule is changed. Intuitively, after comparison time n_i, we expect arms whose means are within $\Delta_i/2$ of the best mean to occupy the first $i-1$ ranks. For $i \in [2:K]$, define

$$\xi_i \triangleq \left\{ \begin{array}{l} \forall u \in \mathbb{N} \text{ with } n_i < u \le n+1,\ \forall k \in [K]: \\ \mu_1 - \mu_k < \dfrac{\Delta_i}{2} \implies \widetilde{\langle k\rangle}_u < i \end{array} \right\}. \tag{12}$$

Set $\xi_{K+1} = \Omega$ and abbreviate $E_i = \bigcap_{p=i+1}^{K+1} \xi_p$. Since only arm 1 satisfies the eligibility condition for ξ_2, and $n_2 = n$, we have $\xi_2 = \{J_n = 1\}$.

Let $\mathcal{F}_{s-1}$ be the information available immediately before drawing I_s, including past actions and observed rewards and the current resolved ranking. These information sets form a filtration by including any next-round tie resolution in $\mathcal{F}_s$, without revealing future action-sampling coins or current rewards before sampling. Thus the ranks and the conditional probabilities $\boldsymbol{p}_{h,s}$ are predictable. The fresh action draw is independent of the current rewards given the past, and the reward observed when arm h is selected has conditional law ν_h. For each arm h, put

$$X_{h,s} = \widetilde{\boldsymbol{g}}_{h,s} - \mu_h, \qquad M_h(t) = \sum_{s=1}^{t} X_{h,s} = \widetilde{G}_{h,t} - t\mu_h, \qquad V_h(t) = \sum_{s=1}^{t} \mathbb{E}[X_{h,s}^2 \mid \mathcal{F}_{s-1}].$$

The importance-weighted estimator and $0 \le Y \le 1$ give

$$\mathbb{E}[X_{h,s} \mid \mathcal{F}_{s-1}] = 0,$$
$$\mathbb{E}[X_{h,s}^2 \mid \mathcal{F}_{s-1}] = \frac{\mathbb{E}_{Y\sim\nu_h}[Y^2]}{\boldsymbol{p}_{h,s}} - \mu_h^2 \le \frac{1}{\boldsymbol{p}_{h,s}}, \qquad |X_{h,s}| \le \frac{1}{\boldsymbol{p}_{h,s}} \le K\overline{\log}K.$$

In particular, M_h is a martingale and V_h is its predictable variance process; no Bernoulli assumption is needed.

Concentration on good-ranking events. We use the following fixed-time Bernstein–Freedman inequality (Freedman, 1975): for martingale differences bounded in absolute value by $B > 0$, their sum M_t and predictable variance V_t satisfy, for deterministic $z, v > 0$,

$$\mathbb{P}(M_t \ge z,\ V_t \le v) \le \exp\left(-\frac{z^2}{2(v + Bz/3)}\right).$$

The same bound holds for $-M_t$. Indeed, for $0 < \lambda < 3/B$,

$$\mathbb{E}[e^{\lambda X_s} \mid \mathcal{F}_{s-1}] \le \exp\left(\frac{\lambda^2 \mathbb{E}[X_s^2 \mid \mathcal{F}_{s-1}]}{2(1 - \lambda B/3)}\right),$$

so $\exp\{\lambda M_t - \lambda^2 V_t/[2(1-\lambda B/3)]\}$ has expectation at most one. Markov's inequality with $\lambda = z/(v + Bz/3) < 3/B$ yields the stated bound. If a good-ranking event E implies $V_t \le v$, then

$$\{M_t \ge z\} \cap E \subseteq \{M_t \ge z,\ V_t \le v\},$$

and hence the same bound applies to its probability. The event E may involve observations after time t or have probability zero: we never condition the stochastic law on E.

Lemma 11 *For the fixed schedule $\boldsymbol{a}$, let $i \in [2 : K]$, $j \in [i : K]$, and let $k \in [K]$ satisfy $\mu_1 - \mu_k < \Delta_i/2$. For every integer t with $n_i \le t \le n$, playing* P1*,*

$$\mathbb{P}\left(\{\widetilde{G}_{k,t} \le \widetilde{G}_{j,t}\} \cap E_i\right) \le 2\exp\left(-\frac{n}{128 H_{\mathtt{P1}}(\boldsymbol{a})}\right).$$

Proof First, the assumptions give $k < i \le j$ and

$$\mu_k - \mu_j = \Delta_j - (\mu_1 - \mu_k) > \Delta_j - \frac{\Delta_i}{2} \ge \frac{\Delta_j}{2} > 0.$$

Splitting a comparison error between the lower tail of the better arm and the upper tail of the worse arm, we obtain

$$\begin{aligned}\mathbb{P}\left(\{\widetilde{G}_{k,t} \le \widetilde{G}_{j,t}\} \cap E_i\right) \le \mathbb{P}\left(E_i \cap \{-M_k(t) > t\Delta_j/4\}\right) \\ + \mathbb{P}\left(E_i \cap \{M_j(t) > t\Delta_j/4\}\right).\end{aligned} \tag{13}$$

Define the doubled-gap rank

$$j^+ = \max\{r \in [j:K] : \Delta_r \le 2\Delta_j\}.$$

The set contains j, so $j \le j^+ \le K$ and $\Delta_{j^+} \le 2\Delta_j$. If $j^+ < K$, then $\Delta_j < \Delta_{j^+ +1}/2$. When $j^+ = K$, the universal rank bound K will suffice. Thus tied gaps and exact doubled-gap equality are included without requiring any additional arm.

We bound both terms in Equation 13 using the same virtual-phase variance calculation. Write

$$S_r = \sum_{\ell=r}^{K} \ell(n_\ell - n_{\ell+1}),$$

and take $(h,r) = (k,i)$ for the better arm and $(h,r) = (j,j^+)$ for the worse arm. In either case $r \ge i$ and $t \ge n_i \ge n_r$. If $r < K$, the corresponding mean-gap condition is $\mu_1 - \mu_h < \Delta_{r+1}/2$: for $h = k$ this follows from $\Delta_i \le \Delta_{i+1}$, and for $h = j$ it follows from the definition of j^+. On E_i, in each block $n_{\ell+1} < s \le n_\ell$, $r \le \ell \le K$, event $\xi_{\ell+1}$ gives $\widetilde{\langle h \rangle}_s \le \ell$ when $\ell < K$; the bound is automatic when $\ell = K$. After n_r, event ξ_{r+1} similarly gives rank at most r, with the universal bound used when $r = K$. All events used have indices larger than i and belong to E_i. Thus good earlier rankings increase sampling probabilities and reduce the conditional variance. Summing the per-round bounds over the disjoint phase blocks and the continuation interval yields, on E_i,

$$\begin{aligned}V_h(t) &\le \overline{\log} K \left[\sum_{\ell=r}^{K} \ell(n_\ell - n_{\ell+1}) + r(t - n_r)\right] \\ &= \overline{\log} K \left[S_r + r(t - n_r)\right].\end{aligned}$$

Empty blocks contribute zero, including when comparison times coincide.

Apply the joint-event concentration bound with $z = t\Delta_r/8$, $B = K\overline{\log}K$, and the positive variance allowance $v = \overline{\log}K\left[S_r + r(t-n_r)\right]$. For either sign separately, it gives

$$\begin{aligned}\mathbb{P}\left(E_i \cap \{\pm M_h(t) \ge t\Delta_r/8\}\right) &\le \exp\left(-\frac{(t\Delta_r/8)^2}{2\overline{\log}K\left[S_r + r(t-n_r)\right] + (K\overline{\log}K)t\Delta_r/12}\right) \\ &= \exp\left(-\frac{t^2\Delta_r^2}{128\overline{\log}K\left[S_r + r(t-n_r) + (K/24)t\Delta_r\right]}\right).\end{aligned}$$

Here the factor 2 multiplies the entire variance allowance; $128 = 8^2 \cdot 2$, and the coefficient $K/24$ comes from $Bz/3$.

The identity $S_r = rn_r + \sum_{\ell=r+1}^{K} n_\ell \ge rn_r$ implies, for $t \ge n_r$,

$$S_r + r(t - n_r) \le \frac{t}{n_r} S_r.$$

Consequently,

$$\frac{t^2\Delta_r^2}{128\overline{\log}K\left[S_r + r(t-n_r) + (K/24)t\Delta_r\right]} \geq \frac{n_r^2\Delta_r^2}{128\overline{\log}K\left[S_r + (K/24)n_r\Delta_r\right]} \geq \frac{n}{128H_{\texttt{P1}}(\boldsymbol{a})}.$$

For the last step, the existing complexity definition, after substituting $n_r = na_r$, reads

$$H_{\texttt{P1}}(\boldsymbol{a}) = n\overline{\log}K \max_{r\in[K]} \frac{S_r + (K/24)n_r\Delta_r}{n_r^2\Delta_r^2}.$$

Finally, $t\Delta_j/4 \geq t\Delta_i/8$ for the better-arm tail, and $t\Delta_j/4 \geq t\Delta_{j^+}/8$ for the worse-arm tail. The two one-sided joint bounds therefore control the two terms in Equation 13; summing them proves the claim for the fixed schedule $\boldsymbol{a}$. ■

Lemma 12 *For the fixed admissible schedule $\boldsymbol{a}$, playing* P1 *in the stochastic setting above, we have for every $i \in [2:K]$,*

$$\mathbb{P}(\xi_i^c \cap E_i) \leq 2K^2 n \exp\left(-\frac{n}{128H_{\texttt{P1}}(\boldsymbol{a})}\right).$$

Proof If ξ_i fails, there are an integer u with $n_i < u \leq n+1$ and an arm k satisfying $\mu_1 - \mu_k < \Delta_i/2$ but $\widetilde{\langle k\rangle}_u \geq i$. Necessarily $k < i$: otherwise $k \geq i \geq 2$ and $\mu_1 - \mu_k = \Delta_k \geq \Delta_i$, contradicting eligibility. At least $i-1$ arms precede k in the estimated ranking, and only $i-2$ other arms can have true labels in $[1:i-1]$. Thus some arm $j \in [i:K]$ precedes k. Since the ranking uses the estimates through round $u-1$, this implies $\widetilde{G}_{k,u-1} \leq \widetilde{G}_{j,u-1}$, including when the estimated scores tie.

Since $n_i \leq u-1 \leq n$, Lemma 11 at $t = u-1$ gives

$$\mathbb{P}\left(\{\widetilde{G}_{k,u-1} \leq \widetilde{G}_{j,u-1}\} \cap E_i\right) \leq 2\exp\left(-\frac{n}{128H_{\texttt{P1}}(\boldsymbol{a})}\right).$$

There are at most $i-1$ eligible arms k, $K-i+1$ competitors j, and $n-n_i+1 \leq n$ ranking times u, since $n_i \geq 1$. A union bound gives

$$\begin{aligned}\mathbb{P}(\xi_i^c \cap E_i) &\leq 2(i-1)(K-i+1)(n-n_i+1)\exp\left(-\frac{n}{128H_{\texttt{P1}}(\boldsymbol{a})}\right)\\ &\leq 2K^2 n\exp\left(-\frac{n}{128H_{\texttt{P1}}(\boldsymbol{a})}\right).\end{aligned}$$

■

Theorem 6 (Upper bounds for P1**)** *For any stochastic problem* STO *with complexity $H_{\texttt{P1}}$ and for any $\boldsymbol{g}$ fixed by an oblivious adversary with complexity $H_{\mathrm{UNIF}}(\boldsymbol{g})$, the probabilities of error of* P1*, denoted $e_{\mathrm{STO}}(n)$ and $e_{\mathrm{ADV}(\boldsymbol{g})}(n)$ in their respective environment, for any n, simultaneously verify*

$$e_{\mathrm{STO}}(n) \leq 2K^3 n\exp\left(-\frac{n}{128H_{\texttt{P1}}}\right) \quad \textit{and} \quad e_{\mathrm{ADV}(\boldsymbol{g})}(n) \leq K\exp\left(-\frac{3n}{40\overline{\log}(K)H_{\mathrm{UNIF}}(\boldsymbol{g})}\right).$$

Proof We consider two cases separately.

The i.i.d. stochastic case For the fixed schedule $\boldsymbol{a}$, the terminal-ranking convention gives

$$e_{\mathrm{STO}}(n) = \mathbb{P}(J_n \neq 1) = \mathbb{P}(\xi_2^c).$$

If ξ_2 fails, take the largest failed index $i \in [2:K]$. Every event with a larger index holds, since ξ_{K+1} is certain. Hence

$$\xi_2^c \subseteq \bigcup_{i=2}^{K} (\xi_i^c \cap E_i).$$

A union bound and Lemma 12 give

$$\begin{aligned} e_{\mathrm{STO}}(n) &\leq \sum_{i=2}^{K} \mathbb{P}(\xi_i^c \cap E_i) \\ &\leq 2(K-1)K^2 n \exp\left(-\frac{n}{128 H_{\mathtt{P1}}(\boldsymbol{a})}\right) \\ &\leq 2K^3 n \exp\left(-\frac{n}{128 H_{\mathtt{P1}}(\boldsymbol{a})}\right). \end{aligned}$$

This holds for every admissible schedule; its virtual comparison times do not affect P1 or the final error probability. The admissible set is finite and nonempty: its entries lie in $\{1/n, \ldots, 1\}$ and the all-ones schedule is allowed. Positive comparison times and gaps make $H_{\mathtt{P1}}(\boldsymbol{a})$ finite and positive. Choosing a minimizing schedule gives

$$e_{\mathrm{STO}}(n) \leq 2K^3 n \exp\left(-\frac{n}{128 H_{\mathtt{P1}}}\right).$$

The adversarial case Fix a deterministic oblivious table $\boldsymbol{g} \in [0,1]^{K \times n}$ and relabel its unique best arm as 1 for analysis. Only the learner is random. It suffices to use the sampling probability guaranteed to every arm; no favorable ranking event is needed. With the information before each action defined as above, the adaptive probabilities satisfy

$$\boldsymbol{p}_{k,t} = \mathbb{P}(I_t = k \mid \mathcal{F}_{t-1}) = \frac{1}{\widetilde{\langle k \rangle}_t \overline{\log} K} \geq \frac{1}{K \overline{\log} K}$$

on every history. Put

$$G_{k,t} = \sum_{s=1}^{t} \boldsymbol{g}_{k,s}, \qquad Y_{k,t} = \widetilde{\boldsymbol{g}}_{k,t} - \boldsymbol{g}_{k,t}, \qquad M_k^{\boldsymbol{g}}(t) = \sum_{s=1}^{t} Y_{k,s} = \widetilde{G}_{k,t} - G_{k,t}.$$

Since the table is deterministic, centering at $\boldsymbol{g}_{k,t}$ is appropriate. Conditionally, the sampling indicator is Bernoulli with parameter $\boldsymbol{p}_{k,t}$, so

$$\begin{aligned} \mathbb{E}[Y_{k,t} \mid \mathcal{F}_{t-1}] &= 0, \\ \mathbb{E}[Y_{k,t}^2 \mid \mathcal{F}_{t-1}] &= \boldsymbol{g}_{k,t}^2 \left(\frac{1}{\boldsymbol{p}_{k,t}} - 1\right) \leq \frac{1}{\boldsymbol{p}_{k,t}} \leq K \overline{\log} K, \\ |Y_{k,t}| &\leq \frac{1}{\boldsymbol{p}_{k,t}} \leq K \overline{\log} K. \end{aligned}$$

Thus $M_k^{\boldsymbol{g}}$ is a martingale with predictable variance

$$V_k^{\boldsymbol{g}}(n) = \sum_{t=1}^{n} \mathbb{E}[Y_{k,t}^2 \mid \mathcal{F}_{t-1}] \le nK\overline{\log}K \quad \text{almost surely.}$$

Recall that $n\Delta_j^{\boldsymbol{g}} = G_{1,n} - G_{j,n}$ for $j \neq 1$ and $\Delta_1^{\boldsymbol{g}} = \min_{j\neq 1} \Delta_j^{\boldsymbol{g}}$. Hence $0 < \Delta_1^{\boldsymbol{g}} \le \Delta_j^{\boldsymbol{g}} \le 1$ and $H_{\text{UNIF}}(\boldsymbol{g}) = K/(\Delta_1^{\boldsymbol{g}})^2$. The error event satisfies

$$\{J_n \neq 1\} \subseteq \{M_1^{\boldsymbol{g}}(n) \le -n\Delta_1^{\boldsymbol{g}}/2\} \cup \bigcup_{j=2}^{K} \{M_j^{\boldsymbol{g}}(n) \ge n\Delta_j^{\boldsymbol{g}}/2\}.$$

Indeed, outside the right-hand side, for every $j \neq 1$,

$$\begin{aligned} \widetilde{G}_{1,n} &> G_{1,n} - n\Delta_1^{\boldsymbol{g}}/2 \ge G_{1,n} - n\Delta_j^{\boldsymbol{g}}/2 \\ &= G_{j,n} + n\Delta_j^{\boldsymbol{g}}/2 > \widetilde{G}_{j,n}. \end{aligned}$$

Thus arm 1 is the unique estimated maximizer, regardless of tie rules.

Apply the Bernstein–Freedman inequality above with $B = K\overline{\log}K$, $v = nK\overline{\log}K$, and $z = n\Delta_k^{\boldsymbol{g}}/2$. The variance bound holds almost surely, so for either sign separately,

$$\begin{aligned} \mathbb{P}\left(\pm M_k^{\boldsymbol{g}}(n) \ge n\Delta_k^{\boldsymbol{g}}/2\right) &\le \exp\left(-\frac{(n\Delta_k^{\boldsymbol{g}}/2)^2}{2[nK\overline{\log}K + (K\overline{\log}K)n\Delta_k^{\boldsymbol{g}}/6]}\right) \\ &= \exp\left(-\frac{n(\Delta_k^{\boldsymbol{g}})^2}{8K\overline{\log}K(1+\Delta_k^{\boldsymbol{g}}/6)}\right) \\ &\le \exp\left(-\frac{3n(\Delta_k^{\boldsymbol{g}})^2}{28K\overline{\log}K}\right), \end{aligned}$$

since $1 + \Delta_k^{\boldsymbol{g}}/6 \le 7/6$. There are K one-sided error events: one lower tail for arm 1 and one upper tail for each other arm. A union bound gives

$$\begin{aligned} e_{\text{ADV}(\boldsymbol{g})}(n) &\le K\exp\left(-\frac{3n(\Delta_1^{\boldsymbol{g}})^2}{28K\overline{\log}K}\right) \\ &= K\exp\left(-\frac{3n}{28\overline{\log}K\,H_{\text{UNIF}}(\boldsymbol{g})}\right) \\ &\le K\exp\left(-\frac{3n}{40\overline{\log}K\,H_{\text{UNIF}}(\boldsymbol{g})}\right), \end{aligned}$$

as claimed. ∎

Appendix G. On the complexities of $H_{\texttt{P1}}$, $H_{\texttt{SR}}$, and H_{BOB}

We now show that for $n \ge \Delta_{(K)}/\Delta_{(1)}$, $H_{\texttt{P1}} = \mathcal{O}\left(H_{\text{BOB}} \log^2 K\right)$. This demonstrates that P1 achieves the best that can be wished for in the two worlds, up to log factors. The extra $\log K$ from the $\log^2 K$ is not always present and we report an even more detailed discussion on the three different regimes of the gaps used in Remark 5 at the end of the section.

Corollary 7 *For $n \geq \Delta_{(K)}/\Delta_{(1)}$, the complexity $H_{\texttt{P1}}$ of* P1 *matches the complexity H_{BOB} from the lower bound of Theorem 4 up to log factors; more precisely,*

$$H_{\texttt{P1}} \leq 2H_{\mathrm{BOB}}(\overline{\log}K)^2 = \mathcal{O}\left(H_{\mathrm{BOB}} \log^2 K\right).$$

For every positive integer horizon n, the stochastic probability of error of P1 *satisfies*

$$e_{\mathrm{STO}}(n) \leq 2K^3 n \exp\left(-\frac{n}{256H_{\mathrm{BOB}}(\overline{\log}K)^2}\right).$$

Proof Recall that $\Delta_{(1)} = \Delta_{(2)}$, and let $Q = \max_{1\leq j\leq K} j/\Delta_{(j)}$, so that $H_{\mathrm{BOB}} = Q/\Delta_{(1)}$. Since the rewards are in $[0,1]$, $0 < \Delta_{(1)} \leq \Delta_{(r)} \leq 1$, $Q \geq 2/\Delta_{(1)}$, and $Q \geq K$. Set the ideal fractions $\alpha_r = \Delta_{(1)}/\Delta_{(r)}$ and the rounded fractions $a_r = \lceil n\alpha_r \rceil/n$ for $r \in [K]$, with $\alpha_{K+1} = a_{K+1} = 0$. Ceilings preserve weak ordering, so $1 = a_1 = a_2 \geq \cdots \geq a_K > 0$ and $1 \leq na_r \leq n$, with $na_r \in \mathbb{N}$. Thus $\boldsymbol{a} \in \boldsymbol{A}$, including when comparison rounds coincide.

For an ordered positive vector b with $b_1 = b_2 = 1$ and $b_{K+1} = 0$, let

$$\Phi(b) = \overline{\log}K \max_{1\leq r\leq K} \frac{\sum_{j=r}^K j(b_j - b_{j+1}) + (K/24)b_r\Delta_{(r)}}{b_r^2\Delta_{(r)}^2}.$$

This is the algebraic expression defining $H_{\texttt{P1}}(\boldsymbol{a})$; we use it for nonintegral comparison times only as an algebraic comparison. The identity

$$\sum_{j=r}^K j(b_j - b_{j+1}) = rb_r + \sum_{j=r+1}^K b_j$$

makes all coefficients in the numerator nonnegative.

Since $\alpha_r\Delta_{(r)} = \Delta_{(1)}$, the second numerator term divided by the denominator is $K/(24\Delta_{(1)}) \leq H_{\mathrm{BOB}}/24$. For the first term, when $r \geq 2$, the identity gives

$$\begin{aligned}\frac{1}{\Delta_{(1)}}\sum_{j=r}^K j(\alpha_j - \alpha_{j+1}) &= \frac{r}{\Delta_{(r)}} + \sum_{j=r+1}^K \frac{1}{\Delta_{(j)}} \\ &\leq Q\left(1 + \sum_{j=r+1}^K \frac{1}{j}\right) \leq Q\left(\overline{\log}K - \frac{1}{2}\right).\end{aligned}$$

For $r = 1$, the weighted sum equals that for $r = 2$ because $\alpha_1 = \alpha_2$. The tail sums are empty when $K = 2$, so this case is also covered. It follows that

$$\Phi(\alpha) \leq H_{\mathrm{BOB}}\overline{\log}K\left(\overline{\log}K - \frac{1}{2} + \frac{1}{24}\right) \leq H_{\mathrm{BOB}}(\overline{\log}K)^2.$$

If $n \geq \Delta_{(K)}/\Delta_{(1)}$, then $n\alpha_r \geq 1$ for every rank, and

$$\alpha_r \leq a_r < \alpha_r + \frac{1}{n} \leq 2\alpha_r.$$

Moving each comparison by less than one round therefore makes the telescoped numerator at most twice its value at α, while the denominator is no smaller. Using the admissible rounded schedule, we obtain

$$H_{\texttt{P1}} \le \Phi(\boldsymbol{a}) \le 2\Phi(\alpha) \le 2H_{\mathrm{BOB}}(\overline{\log}K)^2 = \mathcal{O}\left(H_{\mathrm{BOB}} \log^2 K\right),$$

where the last comparison uses $\overline{\log}K \le 1 + \log K$. Theorem 6 then gives

$$e_{\mathrm{STO}}(n) \le 2K^3 n \exp\left(-\frac{n}{128H_{\texttt{P1}}}\right) \le 2K^3 n \exp\left(-\frac{n}{256H_{\mathrm{BOB}}(\overline{\log}K)^2}\right).$$

For $1 \le n < \Delta_{(K)}/\Delta_{(1)}$, $H_{\mathrm{BOB}} \ge 2/\Delta_{(1)}^2$ gives

$$\frac{n}{256H_{\mathrm{BOB}}(\overline{\log}K)^2} < \frac{\Delta_{(1)}\Delta_{(K)}}{512(\overline{\log}K)^2} \le \frac{1}{512}.$$

Since $2K^3 n \ge 16$ and $\exp(-x) \ge 1 - x$, the displayed error bound exceeds one and follows from $e_{\mathrm{STO}}(n) \le 1$. This proves the error bound for all positive integer horizons, without asserting the complexity comparison for these smaller budgets. ■

G.1. Relation between $H_{\texttt{P1}}$, $H_{\texttt{SR}}$, and H_{BOB} for different regimes of the gaps.

We now study the relation between $H_{\texttt{P1}}$, $H_{\texttt{SR}}$, and H_{BOB} for different regimes of the gaps. We use the same examples as the ones used in Remark 5. We first choose ideal fractions b, with $1 = b_1 = b_2 \ge \cdots \ge b_K > b_{K+1} = 0$, and use the algebraic expression Φ from the preceding corollary proof. Set $a_r = \lceil nb_r \rceil / n$ for $r \in [K]$ and $a_{K+1} = 0$. Ceilings preserve weak ordering and give admissible integer times, even when some coincide; only the analysis changes, while P1 and its horizon are unchanged. If $nb_K \ge 1$, the positive telescoped numerator gives $H_{\texttt{P1}} \le \Phi(\boldsymbol{a}) \le 2\Phi(b)$, as proved above. These schedules compare larger-gap arms earlier, keeping the best two arms at the final comparison time. In the stated horizon ranges, the comparisons are at worst $\mathcal{O}(H_{\mathrm{BOB}} \log^2 K)$, and $\mathcal{O}(H_{\mathrm{BOB}} \log K)$ in the flat and square-root regimes.

▶ **Flat regime** All the gaps are equal, that is, $k \in [2:K]$, we have that $\Delta_k = \Delta_1$.

We choose $b_r = 1, \forall r \in [K]$. All comparisons occur at the final round, and no rounding is needed. The weighted sum equals K at every rank, and $H_{\texttt{SR}} = H_{\mathrm{BOB}} = K/\Delta_{(1)}^2$. Thus the two terms give

$$H_{\texttt{P1}} \le \Phi(b) = \overline{\log}K \left(\frac{K}{\Delta_{(1)}^2} + \frac{K}{24\Delta_{(1)}}\right) \le \frac{25}{24} H_{\mathrm{BOB}} \overline{\log}K,$$

since $\Delta_{(1)} \le 1$. The single-logarithmic comparison holds for every positive integer horizon.

▶ **Super-linear gaps** Since $2 \in \operatorname{argmin}_{r\in[K]}(\Delta_{(r)}/r)$, $\Delta_{(r)} \ge r\Delta_{(1)}/2$ for $r \ge 2$ and $H_{\texttt{SR}} = H_{\mathrm{BOB}} = 2/\Delta_{(1)}^2$. Also, $K\Delta_{(1)} \le 2$ since $\Delta_{(K)} \le 1$.

Let $b_1 = 1$ and $b_r = 2/r$, $r \in [2:K]$. This normalizes the reciprocal-rank schedule, with $b_2 = 1$ and $b_r\Delta_{(r)} \ge \Delta_{(1)}$. The large gaps permit early identification of bad arms. The second complexity term satisfies

$$\frac{K}{24b_r\Delta_{(r)}} \le \frac{K}{24\Delta_{(1)}} \le \frac{H_{\mathrm{BOB}}}{24}.$$

For the first term, telescoping gives, for $r \geq 2$,

$$\sum_{j=r}^{K} j(b_j - b_{j+1}) = 2 + 2\sum_{j=r+1}^{K} \frac{1}{j} \leq 2\left(\overline{\log} K - \frac{1}{2}\right).$$

The rank-1 sum equals the rank-2 sum since $b_1 = b_2$. Using $b_r \Delta_{(r)} \geq \Delta_{(1)}$ gives

$$\Phi(b) \leq H_{\mathrm{BOB}} \overline{\log} K \left(\overline{\log} K - \frac{1}{2} + \frac{1}{24}\right) \leq H_{\mathrm{BOB}} (\overline{\log} K)^2.$$

Since $b_K = 2/K$, rounding gives

$$H_{\mathtt{P1}} \leq 2H_{\mathrm{BOB}} (\overline{\log} K)^2 = \mathcal{O}(H_{\mathrm{BOB}} \log^2 K), \qquad n \geq K/2.$$

▶ **Square-root gaps** We have that $2 \in \operatorname{argmin}_{r \in [K]} (\Delta_{(r)}^2 / r)$, $\sqrt{K/2}\Delta_{(1)} = \Delta_{(K)}$, and also that $\sqrt{r/2}\Delta_{(1)} \geq \Delta_{(r)}$ for each intermediate rank $3 \leq r < K$. The minimizing condition gives the reverse inequality, hence $\Delta_{(r)} = \Delta_{(1)}\sqrt{r/2}$ for every $r \geq 2$. Thus $H_{\mathtt{SR}} = 2/\Delta_{(1)}^2$ and $H_{\mathrm{BOB}} = \sqrt{2K}/\Delta_{(1)}^2 = \sqrt{K/2} H_{\mathtt{SR}}$.

Let $b_1 = 1$ and $b_r = \sqrt{2/r}$, $r \in [2 : K]$. This normalizes the square-root schedule, with $b_2 = 1$ and $b_r \Delta_{(r)} = \Delta_{(1)}$. For $r \geq 2$, the integral comparison $\sum_{j=r+1}^{K} j^{-1/2} \leq 2(\sqrt{K} - \sqrt{r})$ gives

$$\sum_{j=r}^{K} j(b_j - b_{j+1}) = \sqrt{2r} + \sum_{j=r+1}^{K} \sqrt{2/j} \leq 2\sqrt{2K}.$$

The rank-1 sum equals the rank-2 sum. The first complexity term is at most $2H_{\mathrm{BOB}}$, and the second is $K/(24\Delta_{(1)}) \leq H_{\mathrm{BOB}}/24$, since $\Delta_{(K)} = \Delta_{(1)}\sqrt{K/2} \leq 1$. Thus $\Phi(b) \leq (49/24) H_{\mathrm{BOB}} \overline{\log} K$. Since $b_K = \sqrt{2/K}$, rounding gives

$$H_{\mathtt{P1}} \leq 2\Phi(b) \leq \frac{49}{12} H_{\mathrm{BOB}} \overline{\log} K = \mathcal{O}(H_{\mathrm{BOB}} \log K), \qquad n \geq \sqrt{K/2}.$$

For $K = 2$, the tail sums are empty and both normalized schedules reduce to the flat case, without rounding loss. The larger H_{BOB} reflects the extra variance incurred before focusing on the best arms, as discussed in Remark 5. For shorter horizons, these complexity bounds are not asserted. In either nonflat regime the threshold is at most $1/\Delta_{(1)}$ and $H_{\mathrm{BOB}} \geq 2/\Delta_{(1)}^2$, so $n/H_{\mathrm{BOB}} < 1/2$. The corresponding derived probability bounds from Theorem 6 exceed one: the exponent has magnitude less than $1/256$ and $2K^3 n \geq 16$. They follow from $e_{\mathrm{STO}}(n) \leq 1$, as in Corollary 7's proof.